VIGÍA

Multi-Hazard Early Detection for Existing Camera Infrastructure

Technical documentation

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HOW TO READ THIS DOCUMENT

Every quantitative figure is labelled either MEASURED — produced by our own evaluation harness, with the dataset and script named — or REPORTED BY AUTHORS, published by the people who built the component. The two are never mixed, and a figure we have not measured is marked as such rather than omitted. Proportions carry the interval their sample size supports (section 4); several of this project's headline numbers rest on small samples, and a point estimate alone would overstate them.

Numbers in this document are not typed. They are generated from `eval/results/*.json` by `docs/build_paper.py`, so a re-measurement changes the document by rebuilding it.

Abstract

VIGÍA integrates publicly available hazard detectors under a single gated pipeline and adds a shared temporal validation layer that makes them usable on ordinary municipal cameras. The system is explicitly an integration project: every detector it runs was trained by someone else and is credited by name, licence and source. The contribution is the validation layer and the evaluation methodology that measures what that layer buys and what it costs.

The problem is not detection accuracy. Published hazard detectors are good and freely available; what prevents their deployment is that they alarm far too often. Measured on its authors' own validation split, the PyroNear wildfire detector raises an alarm on 0.706 of frames containing no smoke (95 % interval [0.668, 0.741], 585 negatives). That is not a defect but the operating point its authors chose, on the assumption that something downstream filters the output. VIGÍA is that something.

Across 30 negative sequences the validator reduces the false-alarm rate from 0.040 to 0.010, a 75 % reduction, while still detecting every one of the 6 ignition sequences, at a cost of 2.0 minutes of additional latency which this document derives in closed form instead of leaving as an observation. We then run the control most system papers omit — matching that false-alarm rate by simply raising the detector's confidence floor — and report that on this negative set an oracle-tuned floor achieves the same rate at higher recall and lower latency. The validator's defensible value is therefore narrower than raw suppression: it reaches that operating point without access to the negative distribution, and it provides alert deduplication that no threshold can express (section 6.6). The same validator, changed only by configuration, transfers across hazards whose detectors were built by different people, on different data, for phenomena with different physics; a build-failing test enforces that no hazard-specific branch exists in it.

Two commitments are enforced in code, not stated in prose. The runtime imports no AGPL-licensed package, and no biometric identification code path exists; both are checked by guards that parse the source tree and fail the build. A third guard checks every published figure against the results file it cites.

Where the evidence is thin, this document says so and quantifies it. Building access rests on 24 held-out images: its image-level recall is 1.000, but the interval that sample supports is [0.757, 1.000], and the paper reports the second.

Contents

1 Introduction

1.1 Motivation

On 29 October 2024 a DANA — an isolated depression at high levels — struck the province of Valencia, Spain. It killed 223 people, displaced roughly 15,000 residents and caused losses estimated above €50 billion. The Spanish meteorological agency AEMET issued its highest-level red warning hours before the worst of the rainfall, but the regional ES-Alert message to citizens' mobile phones was not sent until late that evening. People died in their homes, in underground garages, and in vehicles overtaken by water.

The gap that day was not a gap in forecasting. It was a gap between what was known and what reached the people affected, and between a hazard beginning and anyone confirming that it had begun. The 2021 eruption of Cumbre Vieja on La Palma, which forced thousands to evacuate over 85 days, showed the same pattern from a different hazard.

Spanish cities already carry tens of thousands of cameras: municipal traffic cameras, road authority cameras, private security systems. During the DANA those cameras were watching the water rise. Nobody was watching the cameras.

1.2 What VIGÍA is

VIGÍA integrates the best publicly available hazard detectors under a single gated pipeline and adds a shared temporal validation layer that makes them operationally trustworthy on ordinary municipal cameras.

The system is an integration project and does not claim novel detection architectures. Every detector it runs was trained by someone else, and each is credited by name, licence and source in section 11. The original contributions are the validation layer of section 6 and the evaluation methodology of section 9, which measures what that layer buys and what it costs.

1.3 The problem the system actually solves

Hazard detection models are widely available and, individually, good. What prevents them from being deployed on municipal cameras is not accuracy but trust: they alarm far too often. Industry reporting puts the proportion of false alarms in security camera systems at roughly 98 %. Commercial wildfire operators compensate with human staffing — Pano AI charges around USD 50,000 per camera per year and runs a 24/7 intelligence centre, and ALERTCalifornia routes detections from more than 1,060 cameras through trained staff before dispatch.

We measured this directly instead of citing it. Run exactly as published, the PyroNear wildfire detector [22] raises an alarm on 0.706 of frames containing no smoke when evaluated against a curated hard-negative set of 585 images spanning 21 cameras (section 5.1). The detector is not defective; that is the operating point its authors deliberately chose, assuming a downstream filter. VIGÍA is that filter.

1.4 Research questions and contributions

This is a measurement-driven systems paper: its purpose is not a new detection architecture but establishing, with controls, what a validation layer over published detectors is worth. Three questions organise it:

RQ1 What does temporal validation buy over the trivial alternative — raising the detector's confidence threshold — and at what cost? (section 6, especially the control in section 6.6.)

RQ2 Does one untuned validation mechanism transfer across hazards with different physics, and can that property be *enforced* rather than asserted? (section 6, section 3.)

RQ3 Which evaluation practices stop a small single-author system from deceiving itself? (section 4, and the corrections recorded throughout.)

The contributions, stated so they can be checked:

1. **A threshold-matched control for alarm-suppression claims** (section 6.6): on easy negatives an oracle-tuned floor dominates our own cascade — a finding we publish against ourselves — while no floor survives curated hard negatives, and deduplication is unreachable by any threshold. We argue this control should be standard in the genre, and it is absent from the systems we compare against.
2. **A measured decomposition of "suppression" into filtering and deduplication** (eq. (15)), which differ by a factor of eight between hazards; single suppression figures conflate two different claims.
3. **A closed-form latency law for persistence**, $\mathbb{E}[\Delta t] = (N-1)/f$, matching measurement exactly and making the recall/latency price of validation predictable before deployment (section 6.7).
4. **Negative results with mechanisms**: two cascade levels removed on evidence — one inverting the safety property on severe floods, one failing because fog is exactly as grey as smoke (section 6.3) — plus a register of the project's own corrected claims.
5. **An enforcement architecture**: licence isolation, a no-biometrics boundary, viewpoint envelopes, figure provenance and the validator's hazard-agnosticism held by build-failing checks, so the paper's claims are continuously re-verified in public.

1.5 Scope and limitations

- VIGÍA is a research prototype, not a certified emergency system. No component has regulatory approval for life-safety use.
- It performs no biometric identification of any kind, by architecture not by configuration (section 3.5).
- Each detector carries a declared operating envelope, enforced before any model is loaded. Claims outside that envelope are not made.
- Several figures come from small samples. Each is reported with the interval its sample size supports (section 4), and the binding cases are listed in section 12.2.
- One hazard — road incidents — is retained as a generality experiment for the validator, not a disaster capability, and has no standalone measurement against held-out labels.

2 Related work

Operational wildfire detection. The systems closest to VIGÍA in purpose run cameras at scale with humans in the loop: Pano AI operates a staffed 24/7 intelligence centre at roughly USD 50,000 per camera per year, and ALERTCalifornia routes detections from more than 1,060 cameras through trained staff before dispatch. PyroNear [22] is the open counterpart — deployed across France, Spain and Chile on Raspberry Pi hardware — and is the direct source of our fire detector. Their training manifest shows sequential confirmation over a minimum frame count, so temporal confirmation *for fire* is established practice; the narrower question this paper measures is whether one untuned mechanism transfers across hazards, and what it buys over the baseline nobody reports against (section 6.6).

Learned smoke detection on the same benchmark. The closest measurement setting to ours is SmokeyNet [6], a spatiotemporal CNN–LSTM–ViT model trained *on FIgLib itself* — the archive our ablation and control experiments draw their sequences from. SmokeyNet attacks the problem from the opposite end: build a better detector by learning temporal structure inside the model. Our work is deliberately complementary: we take detectors *as published* — including by operators who cannot retrain, which is the municipal reality — and measure what an external, model-agnostic validation layer adds. The two approaches compose: a stronger detector lowers the load on the cascade, and the cascade's control experiment (section 6.6) applies to SmokeyNet-class models unchanged. Earlier

CNN pipelines for fire and smoke in surveillance video [20] report strong in-domain accuracy; our negative-set analysis (section 5.1) shows why such figures do not transfer to the false-alarm regime that decides deployability.

Temporal consistency in video object detection. Enforcing agreement across frames is a classical accuracy device in video object detection: Seq-NMS rescores boxes along cross-frame chains [12], tubelet methods link detections into tracks [17], and flow-guided aggregation warps features between frames [31]. Our persistence level shares the association machinery but not the objective: those methods spend temporal structure to *raise mAP* on densely sampled video, whereas the validator spends it to *withhold alarms*, must run on cameras sampling once a minute (where flow and feature aggregation are undefined), and is evaluated on false-alarm rate against detection latency, not accuracy. The distinction matters because the failure modes differ: an accuracy-oriented linker that hallucinates continuity costs a few points of precision; an alarm-oriented one manufactures a confirmed emergency (section 9.4 records exactly this failure on shuffled stills).

Gating and cascades over video. NoScope [16] and CaTDet [19] established that cascaded gating over video buys 5–13× compute at minimal accuracy cost. VIGÍA's Tier 0 applies the same economics declaratively, by registered camera context rather than by learned routing — trading a classifier's silent misrouting failure for a YAML entry a human can audit (section 3.1).

Multi-hazard municipal systems. A recent multi-task system over municipal cameras [1] reports fire false alarms reduced from 52 % to 4 % at 96 % sensitivity using a shared backbone. VIGÍA takes the opposite architectural bet — independent published detectors behind a shared validator — because our hazards arrive from separate datasets with incompatible label spaces, and because a shared backbone couples retraining across hazards (section 3). Their reported Base64-over-WebSocket transport bottleneck also shaped our interface's split transport (section 3.7).

Anticipatory flood sensing. Choi et al. [5] published anticipatory urban flood warning from CCTV using Page–Hinkley change detection — the same capability as section 7, developed independently. We therefore claim no novelty there: we implement both their mechanism and a windowed fit, measure them against each other on identical inputs, and derive a calibration rule (eq. (25)) the measurements bound. V-FloodNet [18] provides the real-camera sequences that comparison runs on.

Flood mapping and anticipation. Deep flood mapping is a mature literature — Bentivoglio et al. [2] review it across satellite, aerial and ground imagery — but it is predominantly *post-event*: knowing what is inundated, not what is filling. The anticipatory gap is the one Choi et al. [5] and our section 7 both target from fixed cameras; our addition is the comparison of two trend mechanisms on identical real-camera inputs, a closed-form false-alarm floor for calibrating one of them (eq. (25)), and a first check of camera-derived coverage against an instrumented gauge (section 7.6).

Datasets and benchmarks. The detectors integrated here rest on ATLANTIS [9], FloodNet [24], SeaDronesSee [28] and DRespNeT [26]; xBD [11] was evaluated and excluded on modality (bi-temporal satellite pairs, not cameras); and the F_2 headline convention follows RipVIS [25]. Every dataset's licence and the exact use made of each appears in section 11.

Position. In genre this is a systems and measurement paper, in the tradition of the gating work above rather than of architecture papers: its claims are quantitative statements about what a mechanism buys, bounded by controls. What separates it from prior systems in this space is that the controls are turned inward — the threshold-matched baseline of section 6.6 is, to our knowledge, absent from every alarm-suppression system we compare against, and running it changed what this paper is allowed to claim. The specific contributions are enumerated in section 1.4.

3 System architecture

VIGÍA is organised in three tiers. Detectors are interchangeable; the validator is shared and hazard-agnostic. Adding a hazard requires a detector class and a configuration entry, and no change to the validator.

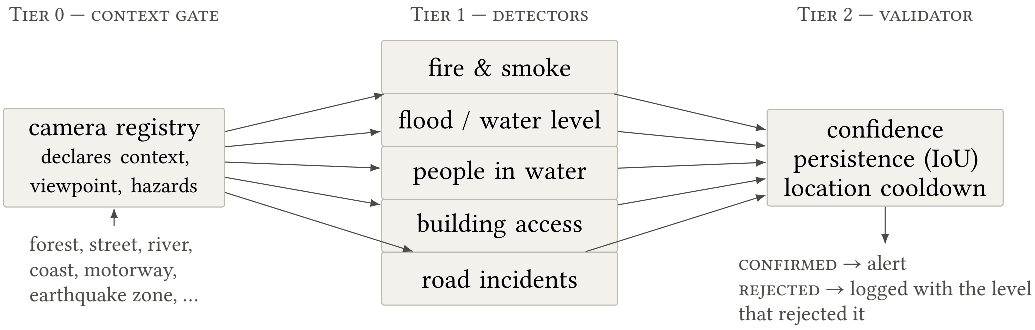

Figure 1. The three tiers. The gate decides which detectors run for a given camera; the detectors run independently and in parallel; one shared validator, configured per hazard but never specialised per hazard, decides which detections become events.

3.1 Tier 0 — the context gate

Each camera is registered once with a declared context. The gate looks up which detectors apply to that context and runs only those.

Definition 3.1 (Gate)**.** Let $\mathscr{X}$ be the set of camera contexts, $\mathscr{H}$ the set of hazards, $\mathscr{C}$ the registered cameras, and

$$\mathscr{V} = \{\,\textsc{ground},\ \textsc{oblique},\ \textsc{nadir}\,\}$$

the set of viewpoints. Each camera c carries a context $\chi(c) \in \mathscr{X}$ and a viewpoint $v(c) \in \mathscr{V}$. The gate is a total map

$$G : \mathscr{X} \longrightarrow 2^{\mathscr{H}},$$

and the detectors constructed for camera c are $\mathscr{D}(c) = \{\,d_h : h \in G(\chi(c))\,\}$.

The gate is declarative, not learned. We deliberately do not classify a frame and route it: that introduces a misrouting failure mode on the critical path and saves little. Every operational deployment we surveyed works this way — a ridgeline wildfire camera never runs drowning detection. Published work on cascaded gating in video pipelines [16, 19] reports 5–13× compute reduction at minimal accuracy cost; here the gate is what makes five hazards affordable on one machine.

The gate's entire intelligence is table 1, which is small enough to audit at a glance. That is the trade being made: a wrong entry here is visible in a YAML file and fixable by a human, whereas a misrouting classifier fails silently.

The saving is reported, not assumed. For a registry of $\mathscr{C}$ cameras and $|\mathscr{H}|$ available hazards, the gate's reduction in detector invocations per frame is

$$\rho \;=\; 1 - \frac{\sum_{c\in\mathscr{C}} |G(\chi(c))|}{|\mathscr{C}|\,|\mathscr{H}|}. \tag{1}$$

On the demonstration registry shipped with the system (8 cameras and $|\mathscr{H}| = 5$ available hazards), the gate requests 8 detector invocations per frame instead of 40, so $\rho = 80\,\%$, and only 4 of the 5

Table 1. The context-to-hazard map. The bias is toward including a hazard when it is plausible, because a missed hazard is a life-safety failure whereas an unnecessary detector costs only the compute the gate was saving. Where that bias was resisted — motorway, which excludes flood — the reason is that claiming a hazard outside a detector's measured envelope is worse than not claiming it.

Context	Hazards enabled	Reasoning
forest	fire	PyroNear's stated envelope: fixed-mount wide-field, daylight
wildland interface	fire, road	Settlement meets vegetation, plus the roads people flee along
street	flood, road, building access	The DANA case: water rising, vehicles caught, people trapped
urban block	flood, building access	Dense built environment
river	flood, drowning	Level and rate of rise, plus anyone swept in
coast	drowning	Open water; the SeaDronesSee envelope
motorway	road	Deliberately narrow: flood is plausible but outside the flood detector's calibrated envelope
earthquake zone	building access	The DRespNeT envelope
UAV search	building access, drowning	Aerial search and rescue; both detectors are UAV-trained

detector types are ever constructed. The unrequested models are never constructed at all, so the saving is in memory as well as compute: a deployment of only ridgeline cameras loads one ONNX session, not five.

3.2 Viewpoint envelopes

The gate also declares *viewpoint*, because a detector's training viewpoint is part of its operating envelope and nothing else in the system captured it.

Definition 3.2 (Admissibility)**.** Each detector d declares an envelope $V(d) \subseteq \mathscr{V}$. A camera c may be paired with d only if

$$\mathrm{adm}(c, d) \iff v(c) \in V(d),$$

and CameraPipeline(c) raises unless $\mathrm{adm}(c, d)$ holds for every $d \in \mathscr{D}(c)$.

This was added after a measured failure, not as a precaution.

A CONFIDENT WRONG NUMBER

The flood segmenter is trained on ATLANTIS [9], which is ground-level and oblique photography of water bodies. Evaluated on straight-down UAV imagery it reported up to 0.397 water coverage on frames whose actual water was a narrow canal, tinting large areas of mown grass as water. It did not fail loudly. It returned a confident wrong number, which for a life-safety component is the worst available failure. Viewpoint is therefore declared on every camera and every detector, and the pipeline refuses the pairing before any model is loaded — the same reasoning as the licence and privacy guards: make the mistake impossible instead of documented. A test fails the build if any detector stops declaring an envelope, so a sixth hazard cannot be added without stating where it works.

FAILING LOUDLY

An unregistered camera raises instead of returning an empty hazard set. Returning an empty set would turn a typo into a camera that watches for nothing and reports success — the worst available failure mode for this system. Enforced by `tests/test_registry.py`.

3.3 Tier 1 — specialist detectors

Independent models running in parallel, rather than one shared-backbone multi-task model. A unified model would require a single coherent label space and jointly annotated data; these hazards come from separate datasets with incompatible labels. It would also couple training, so that retraining fire could silently degrade flood.

3.4 Licensing architecture

VIGÍA ships under Apache-2.0. Several source models were fine-tuned using Ultralytics tooling, which is licensed AGPL-3.0 — a licence whose terms would extend to this entire project were the package imported at runtime.

The resolution is to run every model through ONNX Runtime and never import the `ultralytics` package in the runtime path. PyroNear's own edge engine does exactly this, which is how it remains Apache-2.0 while running a YOLO architecture. Any `.pt` → `.onnx` conversion is an offline build step confined to `tools/`, executed in a separate virtual environment.

This is enforced automatically, not by discipline: `tools/check_licence.py` parses every file under `vigia/` with Python's AST module and fails the build if a forbidden package is imported. It parses rather than greps, so a comment mentioning the package does not trip it.

3.5 Privacy by design

The EU AI Act's high-risk provisions [10] took full effect in August 2026. Annex III classifies biometric identification systems as high-risk, and Article 5(1)(h) prohibits real-time remote biometric identification in publicly accessible spaces for law enforcement, with narrow exceptions.

VIGÍA identifies nobody, and this is structural, not configured.

Table 2. Four privacy commitments and the mechanism that enforces each. A commitment that lives only in a document is a configuration flag waiting to happen.

Commitment	How it is enforced
No biometric code path	`tools/check_privacy.py` parses every file under `vigia/`, `tools/`, `scripts/` and `eval/` and fails on any import of a face or re-identification library, any use of OpenCV's face APIs, or any function or class whose name describes identity work. Unlike the licence guard, `tools/` is not exempt: an offline script that built a face index would breach the commitment just as surely.
Only the event and one evidence frame leave	The `Alert` type has no field for a clip and no writer for one. A test asserts the absence, so adding one becomes a visible change to a documented boundary rather than a quiet tweak.
No raw video retention	`RetentionPolicy` raises on construction if raw video storage is requested. The rolling buffer is a fixed-length in-memory deque sized from the camera's declared retention window, and no code path writes video to disk.
Presence and location, never identity	Person-shaped detections are blurred inside their own boxes before any evidence frame is written, on a copy so the source frame is never mutated. The box stays visible: an operator needs to know that someone is there and where, never who.

The redaction list is keyed on detector class names, and a test fails if a detector declares a person-depicting class that is not on it — so a sixth hazard cannot introduce an unredacted person class by omission.

Guards are tested against violations

A guard that cannot fail is worthless. The privacy guard was run against deliberately planted violations, which is how two gaps were found: class names such as `PersonReidentifier` and `GaitSignatureExtractor` initially escaped, because the patterns matched whole words containing underscores. Matching now strips underscores before comparison. The same reasoning applies to the ONNX parity check of section 9.3.

3.6 Runtime pipeline

The three tiers are joined by `vigia/pipeline.py`: capture, gate, detect in parallel, validate, emit. It decides none of those things itself — the registry decides which detectors run, the validator decides what is an alert, and the egress module decides what may leave. Keeping those three out of the pipeline is what allows each to be tested and reported separately, and it is why adding a sixth hazard touches none of this code.

Concurrency follows the pattern reported for a comparable system [1]: one thread per active detector over bounded queues, with a single fusion stage. The detectors are independent ONNX Runtime sessions, which release the GIL during inference — where essentially all of the time goes — so they run genuinely in parallel. Across cameras the detector objects are shared while the validators are emphatically not: validator state is per camera per hazard, and sharing it would let one camera's tracks confirm another camera's events.

The queues are bounded by design. An unbounded queue converts a detector that cannot keep up into unbounded memory growth and ever-increasing latency, which in a life-safety system means alerting about something that stopped being true minutes ago. Bounded queues convert the same condition into dropped frames that are counted and reported. How the backpressure resolves depends on the source, and the distinction matters for the integrity of every figure here:

- **Live sources never block.** A detector that falls behind costs frames, not latency, and stale frames are discarded in favour of the newest available.
- **File sources always block.** An evaluation replay or a curated demonstration clip must be deterministic; dropping frames would make a published measurement depend on how busy the machine happened to be.

Found by measurement, not foresight

With non-blocking dispatch on both source types, a 16-frame clip through a 193 ms detector processed 3 frames and silently dropped 13. A replay that quietly skips 80 % of its input would have produced plausible and meaningless numbers.

On the reference machine a single ridgeline camera replaying a 40-frame sequence through the fire detector sustains 8–14 frames per second end to end at 117 ms median detector latency, processing every frame with none dropped.

3.7 Operator view

The interface is built around one observation. Every camera-AI demonstration draws a box around a fire, and a reviewer has seen that before. What is rarely shown is detections being *rejected* — and since the validator suppresses the large majority of what the detectors propose, watching that happen is the only direct visual evidence for the one claim in this work that is original. The interface is therefore a dual view: what the detector proposed against what the system confirmed. Every other decision follows from that.

- **Header** — the camera's context and a chip per hazard, lit only for the hazards the gate runs there. Switching cameras visibly changes which detectors are active, which explains Tier 0 faster than a diagram does.
- **Canvas** — confirmed events in accent, candidates still short of the persistence threshold in a muted tone, and suppressed candidates outlined in grey and labelled with the cascade level that rejected them. Naming the level is the point: the viewer sees not only that a detection was dropped but which test dropped it.
- **Signal chain** — the levels of definition 6.1 with live pass and reject counts, the same counters that produce table 10.
- **Timeline** — candidate ticks above confirmed ticks. The density contrast between the two rows is the argument, made without a sentence of text.
- **Metric strip** — seen, confirmed, filtered, deduplicated, frame rate and latency, always visible, in tabular figures so nothing reads as hand-waved. Filtering and deduplication are shown separately here for the reason given in section 6.4.

A single control switches the suppressed layer off, at which point the view becomes an ordinary detection demonstration while the counters keep running. That is the difference between a clean frame and several hundred suppressed candidates behind it: the work is happening either way, and only one of the two views shows it.

The transport is deliberately split. Video is MJPEG over plain HTTP, so the browser's own decoder does the work; events are small JSON over a WebSocket. Frames never travel on the event channel. The reference system [1] identifies Base64-over-WebSocket frame transport as its bottleneck and lists WebRTC as future work; separating the channels sidesteps that problem instead of optimising it. The client falls back to polling if the socket drops, so the view degrades to slower, never to frozen.

The front end is vanilla JavaScript and CSS with no build step and no package manager, and the web framework is an optional dependency rather than a core one — a deployed edge node runs headless and never needs it. Both choices are credibility arguments as much as technical ones: a system that requires a toolchain to demonstrate is a system nobody will try, and an edge runtime of four packages is a claim that can be checked.

A failure that hid behind its own fallback

The WebSocket route rejected every handshake with HTTP 403 while the interface kept updating normally, because the polling fallback covered for it. The cause was that this module used deferred annotation evaluation, which turns annotations into strings that the framework resolves against module globals — while the WebSocket type was imported inside a function, to keep the web framework optional. The parameter could not be typed and was treated as a required query parameter. Worth recording because the graceful degradation that made the system robust is exactly what made the defect invisible; it was found in the server log and a direct client, not in the browser.

Blocking work on the event loop

Several HTTP endpoints were declared asynchronous while doing synchronous work — image decoding, and in one case a complete multi-second analysis of an uploaded clip. An asynchronous handler runs on the event loop, not in a worker thread, so analysing a clip stalled every MJPEG stream and every event socket in the process for the duration, including the live public cameras of section 3.8. The symptom presented as a slow interface rather than as an error. The endpoints that block now run in a thread pool, and the analysis call is dispatched explicitly off the loop; the server answers in milliseconds while an analysis runs.

3.8 Live public cameras

The system runs against genuinely live public cameras as well as recorded clips, through the same pipeline: the United States Geological Survey's streamgage camera network [27], whose imagery is a work of the United States Government and therefore unrestricted, and HPWREN's real-time wildfire cameras [15], which permit publication subject to attribution. Both publish stills every one to five minutes rather than video, and that cadence is the deployment regime, not a limitation of the demonstration — it is precisely why the flood trend detector of section 7 counts samples, not seconds.

Two properties of this path are worth stating because they are easy to get wrong. First, only genuinely new images are admitted: a repeated fetch of an unchanged image is discarded, because handing the validator the same frame three times would satisfy a persistence-of-three test without the world having been observed three times, manufacturing a confirmation from a single observation. Second, no beach or resort webcam appears in the catalogue, and the reason is definition 3.2, not licensing: a beach camera looking along the sand is a GROUND view, while the people-in-water detector's envelope is aerial and elevated. The pairing is refused before a model is loaded.

4 Definitions and evaluation metrics

This section fixes the quantities the rest of the document reports. It exists because several of the mistakes recorded here were mistakes of *metric* and not of model: a figure that was correct as computed and wrong as interpreted. Stating the definitions precisely is what made those visible.

4.1 Matching predictions to ground truth

For two axis-aligned boxes a, b the intersection over union is

$$\mathrm{IoU}(a,b) \;=\; \frac{|a \cap b|}{|a \cup b|} \;\in\; [0,1]. \tag{2}$$

Given predictions $\hat{B} = \{\hat{b}_1, \dots, \hat{b}_m\}$ sorted by descending confidence and ground truth $B = \{b_1, \dots, b_n\}$, each prediction is matched greedily to the highest-IoU unmatched ground-truth box exceeding a threshold τ. Writing TP, FP and FN for the resulting counts,

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{3}$$

Throughout this document $\tau = 0.3$. That is looser than the 0.5 convention, and the choice is deliberate: the operational question is whether a person or a plume was *found*, not whether its box was drawn tightly. A stricter threshold answers a localisation question this system does not ask.

4.2 Two levels of evaluation, and why both

Box-level metrics use eq. (3) over instances. *Image-level* metrics reduce each image to a binary decision — did the detector fire at all — and compute the same quantities over images. The image-level false-alarm rate over a negative set N (images containing no instance of the hazard) is

$$\mathrm{FAR} \;=\; \frac{|\{\, i \in N \,:\, \hat{y}_i = 1 \,\}|}{|N|}. \tag{4}$$

Both are reported everywhere, because either alone conceals a failure the other exposes.

THE FAILURE THAT IMAGE-LEVEL METRICS HID

The people-in-water detector's exported head has six logit slots for five classes. A leading no-object slot was assumed, and every label shifted by one: boats were reported as swimmers at 0.84 confidence while real swimmers were discarded. It nearly survived review because image-level precision still read 0.83 — boats and swimmers co-occur in the same frames, so an image with a boat called "swimmer" is still an image containing a swimmer. Only box-level IoU matching exposed it, at a recall of 0.002.

4.3 Recall-weighted F scores

The harmonic family

$$F_\beta = \frac{(1+\beta^2)\,\mathrm{P}\,\mathrm{R}}{\beta^2\,\mathrm{P}+\mathrm{R}} \tag{5}$$

is reported at $\beta = 2$ as the headline, a convention borrowed from the RipVIS benchmark [25]. The justification is not stylistic. At the symmetric point $\mathrm{P} = \mathrm{R} = p$,

$$\frac{\partial F_\beta/\partial\,\mathrm{R}}{\partial F_\beta/\partial\,\mathrm{P}} = \beta^2, \tag{6}$$

so $\beta = 2$ states explicitly that a point of recall is worth four points of precision here. That is the correct trade for this system for an asymmetric reason: the validator downstream can discard a false alarm, but nothing downstream can recover a person who was never detected.

4.4 Segmentation, and the trap of pooling

For binary masks the water IoU on a single image is eq. (2) over pixel sets. Over a corpus of n images there are two ways to aggregate, and they answer different questions:

$$\mathrm{IoU}_{\mathrm{pooled}} = \frac{\sum_{i=1}^{n} |P_i \cap G_i|}{\sum_{i=1}^{n} |P_i \cup G_i|}, \tag{7}$$

$$\mathrm{IoU}_{\mathrm{strat}} = \frac{\sum_{i\in S} |P_i \cap G_i|}{\sum_{i\in S} |P_i \cup G_i|}, \qquad S \subset \{1,\dots,n\}. \tag{8}$$

Equation (7) weights every image by the size of its union, so a corpus dominated by easy cases is dominated by easy cases in the score. When the subset S that the model exists to handle is a small minority, the pooled figure flatters it, and the more severe the case the smaller its influence.

This is not hypothetical. FloodNet's [24] labelled release contains 398 pairs of which only 51 are flooded — 12.8 % of the corpus. The aerial flood segmenter scores $\mathrm{IoU}_{\mathrm{pooled}} = 0.7021$ but $\mathrm{IoU}_{\mathrm{strat}} = 0.4734$ on the flooded subset. Model selection in this project uses the stratified figure, and section 5.2 publishes it as the headline with the pooled one beside it, clearly labelled. Selecting on the pooled figure would have chosen a model that is better at recognising dry ground.

4.5 Interval estimation

Half of this project's figures rest on small samples: 24 held-out images for building access, 23 negatives in the first fire baseline. A bare point estimate from such a sample invites a reader to believe it to three decimal places, so every proportion in this document whose denominator is known is reported with a 95 % *Wilson score* interval [29]. For k successes in n trials with $\hat{p} = k/n$ and $z = 1.96$,

$$\mathrm{CI}_{95} = \frac{\hat{p} + \dfrac{z^2}{2n}}{1 + \dfrac{z^2}{n}} \pm \frac{z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}. \tag{9}$$

Wilson over the textbook normal approximation $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$, for a reason this project's data makes concrete instead of academic [3]. At $\hat{p} = 1$ the normal interval collapses to a point:

building access detects a person in 12 of 12 positive held-out images, and the normal interval reports $[1.000, 1.000]$ — perfect certainty inferred from twelve observations. Equation (9), derived by inverting the score test rather than by assuming normality of the estimate, gives [0.757, 1.000]. That interval is the honest statement, and it is what section 5.4 reports.

Table 3 collects the cases where the interval changes how a figure should be read.

Table 3. Point estimates with the 95 % Wilson interval their sample size supports. The last column is the reading the interval permits, which in three of these rows is materially weaker than the point estimate alone suggests. Computed by `docs/build_paper.py`, not typed.

Quantity	Estimate	95 % interval	What it supports
Fire image-level FAR, 585 negatives	0.706	[0.668, 0.741]	Precise enough to quote. The motivating claim is safe.
Fire image-level FAR, 23 negatives	0.870	[0.679, 0.955]	Consistent with anything from two-thirds to nearly all. Withdrawn.
Swimmer box recall, 1218 instances	0.9146	[0.8976, 0.9290]	Tight. The strongest evidence in the project.
Civilian box recall, 291 instances	0.942	[0.908, 0.963]	Solid at instance level, from 24 images.
Building image-level recall, 12 images	1.000	[0.757, 1.000]	Cannot exclude one image in four being missed.
Building image-level FAR, 12 negatives	0.083	[0.015, 0.354]	Almost uninformative. Reported for completeness only.
Scene router accuracy, 384 images	1.000	[0.990, 1.000]	Tight, but see section 8 on task difficulty.

4.6 Class merging

Two detectors emit more classes than VIGÍA consumes. Merging is a partial surjection

$$\mu : \mathscr{K}_{\text{model}} \longrightarrow \mathscr{K}_{\text{system}} \cup \{\perp\}, \tag{10}$$

where $\perp$ discards a class. For building access $|\mathscr{K}_{\text{model}}| = 28$ and $|\mathscr{K}_{\text{system}}| = 5$; for people in water, five model classes reduce to one consumed class. Crucially, μ is applied at the detector boundary and only a designated subset $\mathscr{A} \subseteq \mathscr{K}_{\text{system}}$ — the *alarm* classes — reaches the validator. Everything else is operator context. A rescue team on site is the expected state at an incident already being handled, and a boat is not an emergency; routing either to the validator would alarm on every rescue worker and every vessel in frame.

5 Detectors

One subsection per hazard. Each names the people whose work it uses, states what we changed, and reports only figures we measured ourselves; figures published by the original authors are listed separately and attributed. Figure 2 shows one annotated frame per hazard, drawn by the same overlay the running system uses.

Five hazards are integrated. Four are disaster hazards and carry the system's purpose; the fifth, road incidents, is retained explicitly as a generality experiment for the validator, not a headline capability — a car accident is an emergency, not a disaster. Its value is that it has completely different physics and timescales from wildfire, so a validator that works on both is demonstrably not tuned to one phenomenon.

5.1 Fire and smoke

Model `yolo11s_sensitive-detector`, Apache-2.0.

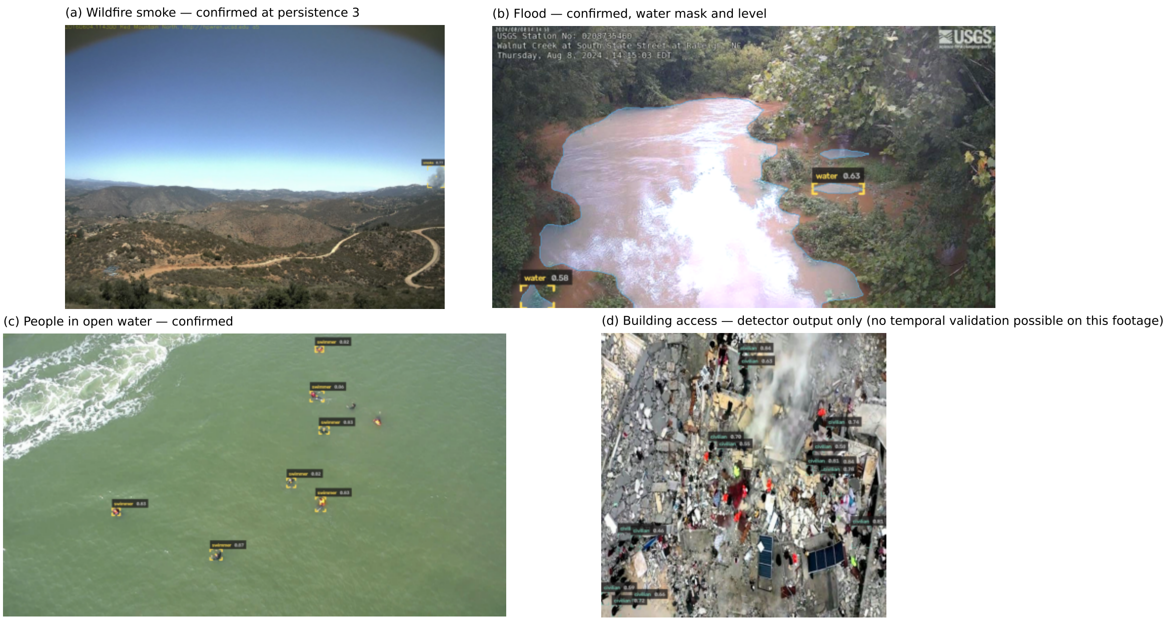

Figure 2. Qualitative results, one frame per hazard, rendered by `scripts/make_qualitative_figure.py` under the same licence gating as every published artefact. (a) Wildfire smoke confirmed at persistence 3 on a HPWREN FIgLib ignition sequence (HPWREN, UC San Diego — attribution required). (b) Flood: the segmentation mask with its boundary, and the confirmed regions, on a USGS streamgage camera during the August 2024 Walnut Creek flood (public domain). (c) Seven people in open water confirmed simultaneously (SeaDronesSee, CC0). (d) Building access is *detector output only*, shown at confidence ≥ 0.45 for legibility — its released footage has no temporal continuity, so no confirmed state exists to show (section 9.4); DRespNeT, CC BY 4.0, UAV imagery of the 2023 Türkiye earthquakes.

Authors	The PyroNear association, a French open-source non-profit, in collaboration with the French Fire and Rescue Services (SDIS) [22].
Deployment	15 lookout towers, 51 cameras across France, Spain and Chile, running on Raspberry Pi hardware.
Envelope	Fixed-mount wide-field outdoor cameras viewing a landscape horizon, in daylight. Not validated for indoor fire, close-range flame or night operation.

It is a deliberately *sensitive* detector: the authors recommend a low confidence floor (0.20) and a very low NMS IoU (0.01), trading precision for recall on the assumption that a downstream filter exists. That design choice is precisely why it is the right baseline for this project — the filter it assumes is what we built.

We run the published ONNX export unmodified. The model is downloaded from its original distributor at setup time and is never redistributed here. We relabelled its single class from `item` to `smoke`, and wrote our own letterbox preprocessing and YOLO decoding with NumPy non-maximum suppression, so the runtime has no dependency on the AGPL-licensed package (section 3.4).

The authors publish no precision, recall or mAP for this checkpoint, so table 4 has no REPORTED BY AUTHORS column: there is nothing to compare against, and inventing one would be worse than the gap.

A caveat that earned its keep

The most quoted figure in this project — that the published fire detector alarms on frames containing no smoke — was first measured on 23 negative frames and carried the standing note "indicative, not precise; widen before publishing". Widening it to 585 negatives from 21 cameras, a twenty-five-fold increase, moved the false-alarm rate from 0.870 to 0.706 and box precision from 0.625 to 0.447. Recall barely moved (0.904 to 0.902), which is the reassuring half: the detector finds what it finds, and it was the small sample that had flattered it on false

Table 4. Fire detector, single-frame and with no temporal validation — this is the baseline the validator must improve on, not a headline result. `pyro-sdis` [23] validation split, 1085 rows (500 positive, 585 negative, 21 cameras), confidence 0.20, $\tau = 0.3$. MEASURED — `eval/run_image_eval.py`.

Metric	Value	Note
Box precision	0.447	
Box recall	0.902	
Box F_2	0.750	recall-weighted, eq. (5)
Image-level recall	0.978	
Image-level FAR	0.706	95 % CI [0.668, 0.741]
Median latency, ms	72.5	M4 Pro, CPU provider

alarms. The motivating claim is untouched — a detector alarming on 71 % of clear frames is unusable without a filter — but the specific number was overstated by sixteen percentage points and every quotation of it was corrected. Table 3 shows why the original was never quotable: its interval was [0.679, 0.955].

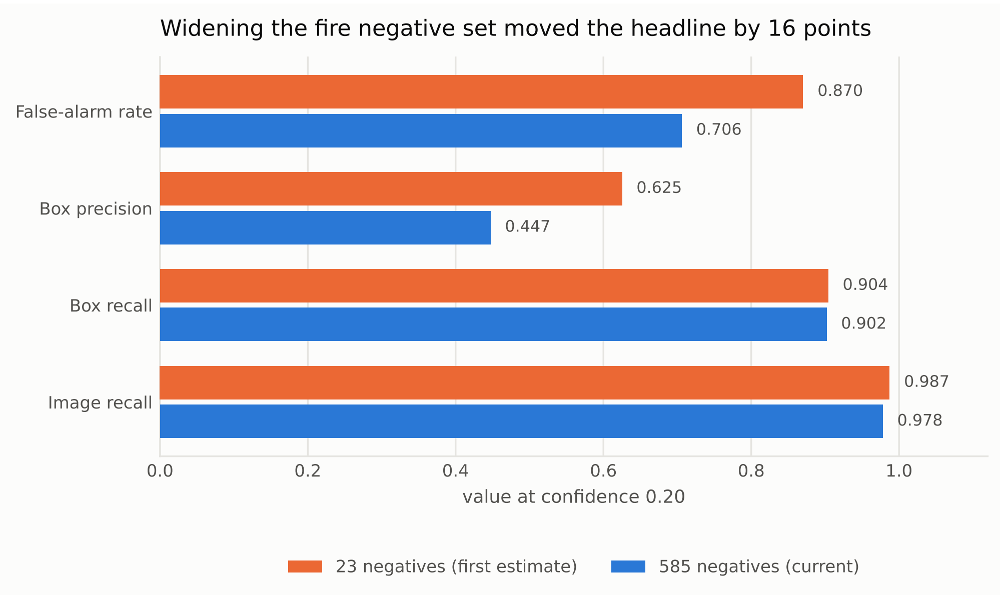

Figure 3. The same detector at the same threshold, measured against 23 negatives and then against 585. Recall barely moved; precision and the false-alarm rate both fell. This is what a small sample flattering a detector looks like.

5.2 Flood and water level

The keystone hazard, and the one directly connected to the DANA. Two capabilities: segmenting the water region, and estimating water level against a fixed per-camera reference line (section 7).

Architecture	DeepLabV3 [4] with a MobileNetV3-Large backbone [14], from torchvision (BSD-3-Clause).
Training data	ATLANTIS [9] — 5,195 pixel-annotated waterbody images, 56 classes.
Trained by	Us. This is the only ground-level detector in VIGÍA that we trained ourselves.
Envelope	Water region segmentation in outdoor scenes, ground and oblique viewpoints. The per-camera reference line is optional: without calibration the uncalibrated area signal still yields a trend, so a camera is useful the moment it is connected.

The architecture was chosen for licensing as much as performance: torchvision is BSD-3-Clause, making this the only detector in the system entirely free of AGPL. ATLANTIS's 56 classes were collapsed to binary water by eq. (10) — VIGÍA does not need to know whether it is looking at a canal or a river, only where the water is and whether it is rising. Seventeen liquid-water labels count as water; water-associated *structures* (dam, levee, pier, culvert, breakwater) are excluded because they are not water, as are glaciers and snow, which are not liquid, and mangrove, which is vegetation standing in water whose extent does not track a level. The class mapping was derived empirically by cross-referencing the dominant mask value in each label folder against the published label list, not assumed.

Table 5. Flood segmenter on the ATLANTIS test split, 1296 images, held out and never seen during training or validation. No third-party baseline is quoted because the dataset authors' own network reports over all 56 classes rather than binary water, so its figures are not comparable. MEASURED — `eval/run_flood_eval.py`.

Metric	Value	Note
Water IoU	0.7633	pooled, eq. (7)
Precision	0.8799	
Recall	0.8520	
F_1	0.8657	
Pixel accuracy	0.9319	
Median latency, ms	21.8	M4 Pro, CPU provider

A DEPLOYMENT HAZARD IN THE EXPORT

PyTorch's ONNX exporter splits weights into a separate `.onnx.data` sidecar. Copying only the `.onnx` yields a graph that loads successfully and then produces garbage. Our export folds any sidecar back into the single file and verifies argmax parity against the source PyTorch model, failing below 99.9 % agreement.

5.2.1 The aerial variant

A UAV camera declaring a NADIR viewpoint was previously refused by definition 3.2, because the ATLANTIS model's envelope excludes it. A second segmenter was therefore trained on FloodNet Track 1 [24] (CDLA-Permissive-1.0) and the pipeline routes by viewpoint.

Its evidence class is weaker than everything else in this document and is labelled as such: FloodNet withheld its validation and test masks for the competition, so no independent labelled split exists in the release. The figures below come from a stratified split carved out of the 398 labelled training pairs and *seen by model selection* — training-time validation, not held-out evaluation.

Table 6. Aerial flood segmenter. Two IoU figures are reported because section 4.4 requires it: only 51 of 398 labelled images are flooded, so the pooled figure is carried by dry scenes. Model selection used the stratified figure. VALIDATION-MEASURED, not held out — `scripts/train_flood_aerial.py`.

Metric	Value	Note
Water IoU, flooded subset	0.4734	eq. (8); the honest figure
Water IoU, pooled	0.7021	eq. (7); flattered by dry scenes
Pixel accuracy	0.9572	
Best epoch	25	selected on the flooded subset

5.3 People in water

Model RF-DETR Small fine-tuned on SeaDronesSee [7], Apache-2.0.

Dataset SeaDronesSee [28], CC0 1.0 — the least restrictive licence in the project, so the credit here is given by choice rather than obligation.

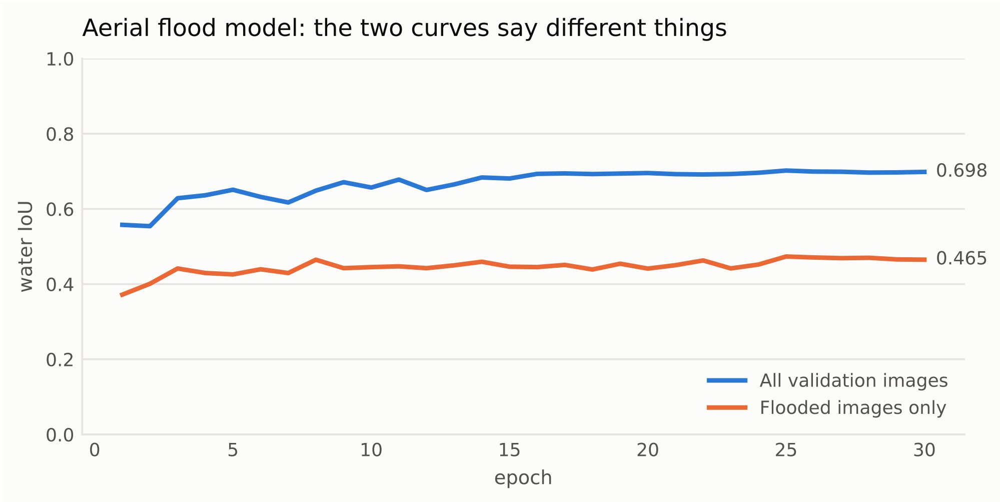

Figure 4. Training the aerial flood segmenter. The two curves are reported separately because only 51 of 398 labelled images are flooded; a single pooled IoU would be carried by dry scenes and would flatter the model on the one case it exists to handle.

Why RF-DETR The same author published YOLO11n and YOLO11x checkpoints for this dataset under AGPL-3.0. RF-DETR was chosen specifically because it is Apache-2.0 and keeps this detector outside the AGPL perimeter.

Envelope Elevated and aerial views of open water, 5 m to 260 m altitude. Not validated for pool settings, surf zones or extreme crowd density.

Only the `swimmer` class reaches the validator. A boat is context, not an emergency. RF-DETR emits a fixed set of object queries with no duplicate boxes, so unlike the YOLO detectors it requires no non-maximum suppression.

Table 7. People-in-water detector, SeaDronesSee validation split, 300 images (255 with swimmers), 1218 ground-truth swimmer instances, confidence 0.30. MEASURED — `eval/run_person_in_water_eval.py`.

Metric	Value	Note
Swimmer box precision	0.9199	
Swimmer box recall	0.9146	95 % CI [0.8976, 0.9290]
Swimmer box F_1	0.9172	
Image-level FAR	0.0222	95 % CI [0.0039, 0.1157]
Median latency, ms	50.2	M4 Pro, CPU provider

We do not quote the author's headline mAP@50 of 0.7931 (REPORTED BY AUTHORS): it is across five classes and carried by boats, and the swimmer class — the only one VIGÍA cares about — is their second weakest at 0.282 (REPORTED BY AUTHORS). Our 0.9172 and their 0.282 are not in conflict. Average precision over strict IoU thresholds punishes imprecise boxes on tiny objects, while our $\tau = 0.3$ operating point asks the operationally relevant question, which is whether the person was found at all.

5.4 Trapped people and building access

Model YOLOv8-DRN, trained and published by the DRespNeT authors [26], CC BY 4.0 — attribution is legally required here, unlike SeaDronesSee where it is given by choice.

Dataset DRespNeT: UAV imagery of the 2023 Türkiye earthquakes, 650 train / 50 valid / 25 test.

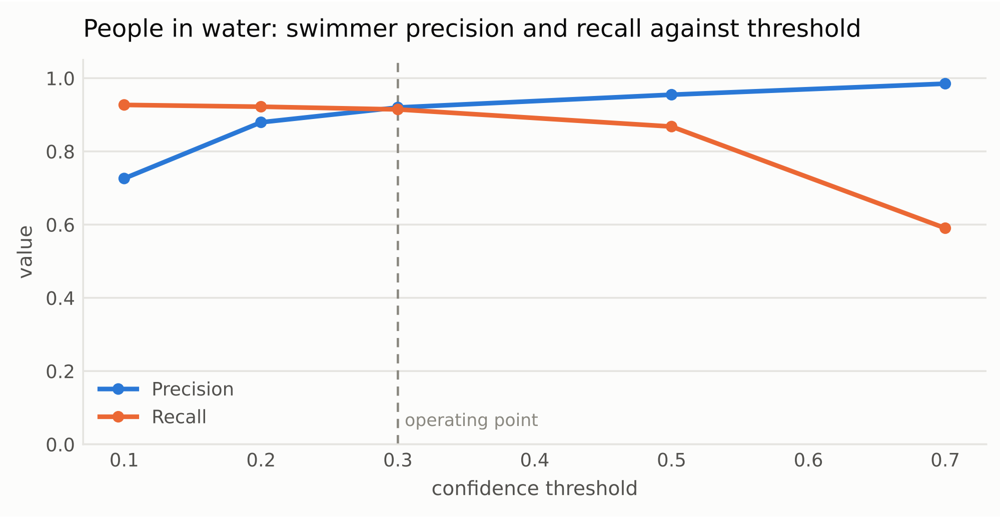

Figure 5. Swimmer precision and recall against the confidence threshold. The operating point is chosen where recall is still high, because the validator can discard a false alarm but nothing downstream can recover a person never detected — the same asymmetry that motivates $\beta = 2$ in eq. (6).

Envelope Downward and oblique UAV views of earthquake-damaged urban blocks. Not validated for street-level views, interior imagery or night. Trained on 650 images, so generalisation to other disaster types and other cities is unproven and treated as unproven.

The model predicts 28 classes; eq. (10) reduces them to five at the detector boundary, and only `civilian` reaches the validator. The model is an instance segmenter, so its output carries 32 mask coefficients after the class scores; these are trimmed at postprocess, because without that the decoder reads 60 classes and takes an argmax over mask prototypes.

Table 8. Building-access detector on the held-out test split, 24 images, 291 ground-truth civilian instances, confidence 0.30. The image-level rows are computed over twelve positive and twelve negative images; their intervals are given because the point estimates alone would badly overstate what 24 images can establish. MEASURED — `eval/run_building_access_eval.py`.

Metric	Value	Note
Civilian box precision	0.785	
Civilian box recall	0.942	95 % CI [0.908, 0.963]
Civilian box F_1	0.856	
Image-level recall	1.000	95 % CI [0.757, 1.000]
Image-level FAR	0.083	95 % CI [0.015, 0.354]
Rescue team F_1	0.895	operator context, not an alarm
Accessible entry F_1	0.882	operator context
Blocked entry F_1	0.802	operator context
Collapsed building F_1	0.941	operator context
Median latency, ms	37.6	M4 Pro, CPU provider

The authors report 92.7 % mAP@50 at 27 FPS across all 28 classes (REPORTED BY AUTHORS). That is not comparable to table 8: they report over 28 classes, we merge to five and measure one.

An architectural preference abandoned on evidence

VIGÍA first trained its own torchvision Faster R-CNN on the same 650 images, specifically because torchvision is BSD-3-Clause and would have kept this detector clear of the AGPL perimeter altogether. On the identical held-out split with the identical metric it reached civilian box recall of 0.141 — it missed roughly six of every seven people it should have boxed, and was by a wide margin the weakest component in the system. The authors' own weights reach 0.942. The test/valid gap also inverted: our model scored 0.462 on valid against 0.141 on test, a collapse showing it had not generalised, where the published weights score 0.815 against 0.856. Keeping a six-times-worse detector to preserve an architectural preference would have been a choice against the people this detector exists to find. The AGPL exposure is contained the same way as fire: ONNX conversion in an isolated build environment, no weights redistributed, enforced by `tools/check_licence.py`.

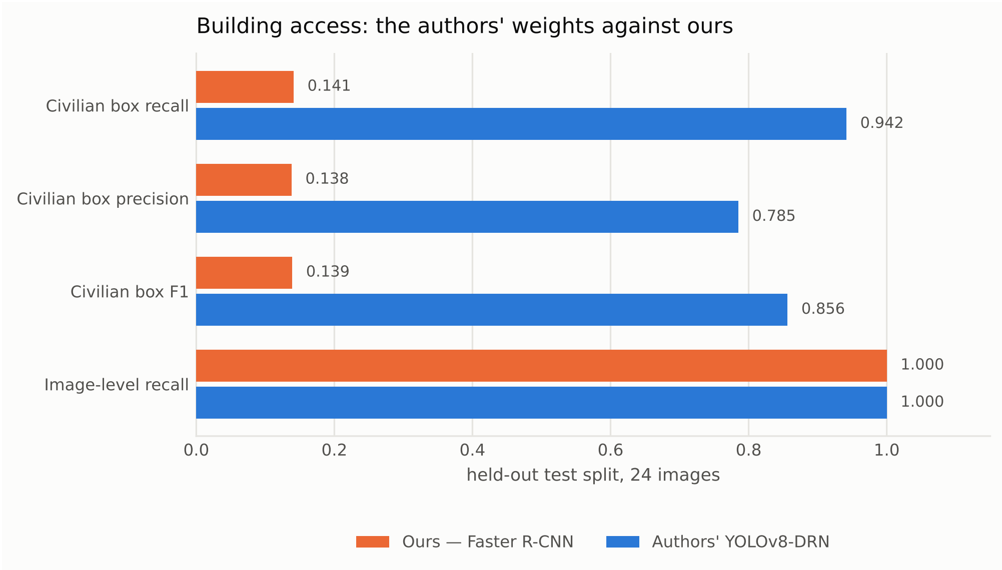

Figure 6. The two models on the identical held-out split with the identical metric. Image-level recall is 1.000 for both, which is exactly why image-level metrics alone are not enough (section 4.2): they hide a six-fold difference in whether the person was actually located.

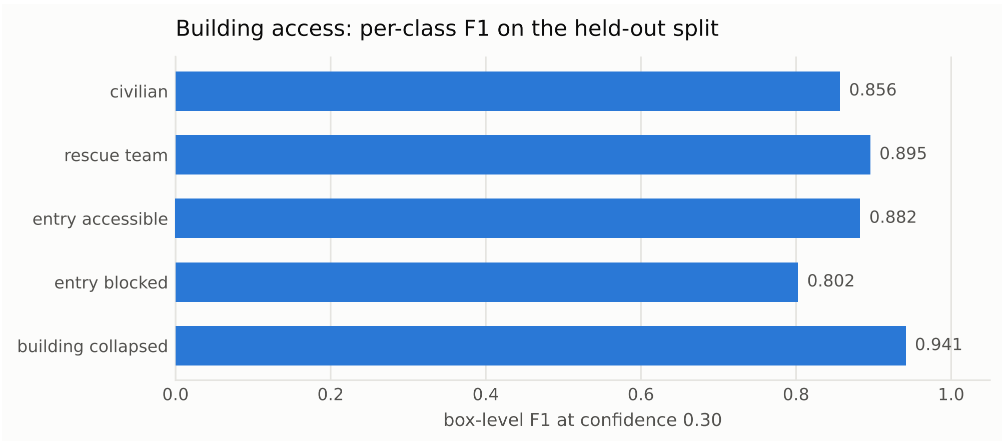

Figure 7. Per-class F_1 on the held-out split at confidence 0.30. Only `civilian` reaches the validator; the rest are operator context.

5.5 Road incidents

Included to test whether the validator transfers to a hazard with entirely different physics. The model [8] is MIT-licensed and detects accidents and vehicles in roadside and dashcam imagery; only the `accident` class reaches the validator, since passing `vehicle` through would make every car on the road an alert.

Its status is *not measured.* It is integrated and running behind the validator, and it supplied the second hazard in table 11, but that experiment measures the *validator*, not this detector: no precision or recall for the accident class against held-out labels has been run. The available footage is a dashcam compilation that is predominantly accident footage and therefore cannot support a false-positive rate. The authors report mAP@50 of 0.826 and accident-class recall of 0.855 (REPORTED BY AUTHORS); we do not adopt those as ours.

A CHECKPOINT CHOICE RECORDED, NOT SILENTLY MADE

The author's own inference script uses the `epoch14` checkpoint. On their four published test images, `epoch61` returns exactly one accident per image at confidence 0.58–0.70, whereas `epoch14` returns three detections on one image. We use `epoch61` and record the reason here so a future reader need not wonder why the two disagree.

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6 The temporal validator

This is the original contribution. Everything else in VIGÍA is integration.

6.1 Rationale

Detection is a commodity: datasets are public, models are published, and fine-tuning one is a weekend's work. The unsolved, expensive and operationally decisive problem is the false alarm, which is why commercial operators surround good models with expensive human staffing.

The validator sits between the detectors and the alert channel. It receives every raw detection and decides which become events. Detections it rejects are not discarded silently: each is annotated with the name of the level that rejected it, so suppression is auditable in evaluation and visible in the operator interface.

6.2 The cascade

Definition 6.1 (Cascade)**.** Let x be a detection with box $b(x)$, confidence $s(x)$ and frame timestamp t. The validator applies three predicates in order, cheapest first:

$$L_1(x) \iff s(x) \geq \theta \qquad \text{confidence} \tag{11}$$

$$L_2(x) \iff h(\kappa(x)) \geq N \qquad \text{persistence} \tag{12}$$

$$L_3(x) \iff \neg\,\exists\,(b',t') \in R_t \,:\, \mathrm{IoU}(b(x),b') \geq \tau_c \qquad \text{cooldown} \tag{13}$$

where $\kappa(x)$ is the track x was associated with, $h(\cdot)$ is that track's hit count, and $R_t = \{(b',t') \,:\, t-t' \leq \Delta\}$ is the set of recently confirmed boxes. A detection becomes an event iff $\bigwedge_{i=1}^{3} L_i(x)$.

Association is by greedy IoU overlap: a detection joins the existing track maximising IoU above a threshold τ_t, or starts a new one, and a track is dropped after M consecutive misses. Every level is independently switchable. That is not a convenience — it is what allows the ablation of section 6.5 to be produced by the same code that runs in production, and not by a separate analysis script that might diverge from it.

6.3 The cascade earned its shape: two levels built, measured, removed

The cascade did not start with three levels. Two more were implemented, measured under the same harnesses, and removed on the evidence; both removals are guarded by regression tests, and both are reported here because a cascade that only ever grows is a cascade nobody is checking.

Size plausibility: harmful on the worst cases

A size-plausibility level rejected detections whose area fraction or aspect ratio was implausible. Measured, it rejected 0 % of detections on wildfire and 0 % on road incidents. When flood was added it proved actively harmful: real ATLANTIS flood masks cover a median 32 % of the frame and severe cases exceed 60 %, so a plausible maximum-area default silently discarded the worst floods — the more severe the disaster, the more likely the alert disappeared. A level that earns nothing on two hazards and inverts the safety property on a third does not belong in a life-safety cascade. Removing it changed no measured result, which is itself the evidence it was doing nothing.

Colour prior: fog is exactly as grey as smoke

A colour level checked the fraction of a box's pixels inside a per-hazard HSV region. Every shipped configuration disabled it, which meant an earlier draft of this paper defined a level that had never run in a reported experiment — the same situation that triggered the size measurement. The only physically sensible prior for wildfire smoke is achromaticity, and as first parameterised the level could not even express that (it could demand *minimum* saturation, never a maximum); the bound was added so the measurement could run at all. Swept over five cutoffs on 1085 pyro-sdis stills, the prior bought false alarms only by discarding real smoke — recall 0.978 → 0.736 to move FAR 0.706 → 0.456 at the strictest cutoff, about five points of recall per ten of FAR at every point — because fog, the curated hard negative, shares the one property being tested. Inside the temporal cascade it improved nothing at any cutoff, and strict cutoffs lost an ignition sequence outright (5 of 6). Removed (`eval/run_colour_eval.py`; the harness reimplements the prior locally so the experiment outlives the level it deleted).

6.4 Filtering and deduplication are different claims

The validator rejects the large majority of what the detectors propose, and it is tempting to report that as a single suppression figure. That figure would be misleading, because it combines two operations supporting entirely different claims.

Definition 6.2 (Suppression decomposition)**.** Let P be the number of proposals and E the number of confirmed events. Write Φ for the proposals rejected by L_1–L_2 and D for those rejected by L_3, and define

$$S_{\text{total}} = 1 - \frac{E}{P}, \qquad S_{\text{filter}} = \frac{\Phi}{P}, \qquad S_{\text{dedup}} = \frac{D}{P}. \tag{14}$$

Since every proposal is either confirmed or rejected by exactly one level,

$$S_{\text{total}} = S_{\text{filter}} + S_{\text{dedup}}. \tag{15}$$

Only the first term is false-alarm reduction, and only the first term supports the claim this project makes. The second is the system declining to re-announce an event it has already reported: nothing is filtered out there, an alert is simply not sent twice.

Measured across the four demonstration clips, 1255 proposals produced 117 confirmed events. As one number that is 91 % suppression. Split by eq. (15) it is 74.4 % filtering and 16.3 % deduplication — and the split varies enormously by hazard.

This project has been caught by exactly this before

During the first ablation, recall appeared to collapse from 0.925 to 0.125 and read as the validator destroying the detector. It was cooldown correctly suppressing repeat alerts about a fire already found — the validator was locating *more* fires, not fewer. Cooldown was removed from the evaluation harness for that reason. The

Table 9. The decomposition of eq. (15) per hazard. The proportions invert between fire and building access, which a single suppression figure would conceal entirely. MEASURED — `docs/figures/suppression.json`.

Hazard	Proposed	Confirmed	Filtering %	Dedup. %
Fire	20	2	10.0	80.0
Flood	61	8	23.0	63.9
People in water	451	59	70.7	16.2
Building access	723	48	82.8	10.5

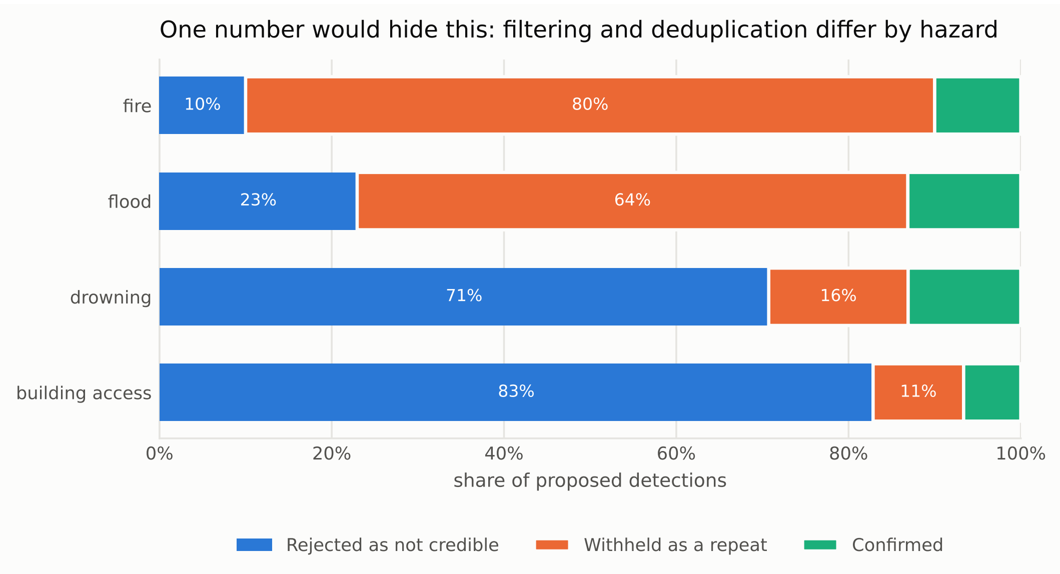

Figure 8. What the validator does with each hazard's proposals. Fire's headline suppression is almost entirely the system declining to re-announce one fire it had already found; building access's is almost entirely filtering. Quoting a single figure would overstate the false-alarm contribution by a factor of eight on the fire clip.

runtime pipeline later reintroduced the same conflation in its live statistics, and now reports the two quantities separately everywhere they appear, including on the recorded demonstrations and in the operator interface.

6.5 Does it reduce false alarms?

Table 10. The validator against the raw detector on 30 negative sequences and 6 ignition sequences from HPWREN FIgLib. MEASURED — `eval/run_ablation.py`.

Configuration	FAR	Reduction %	Fires found	Frame recall	Min. to detect
Raw detector, no validator	0.040	—	6/6	0.925	1.0
Full cascade, persistence = 3	0.010	75	6/6	0.792	3.0

At persistence 3 the false-alarm rate falls by 75 % and every fire is still found: 6 of 6, the same as the raw detector. Sequence-level detection is what matters operationally, and it is unchanged. A ByteTrack-style second association stage [30] — below-floor detections keeping tracks alive without being allowed to alarm — was also implemented and measured: it changed nothing on either the sparse ablation or a dense 8 fps sequence, because the tracker's miss tolerance already bridges the gaps it exists to bridge. It ships disabled, as a documented null result.

Two costs are paid for that, and both sit in table 10, not in a footnote. The median time to first alert rises from 1.0 to 3.0 minutes, and frame-level recall falls from 0.925 to 0.792 — the frames spent accumulating persistence are frames on which the system stays silent about a fire it can already see. Reporting only the false-alarm reduction would hide both. A filter that drove false alarms to zero by refusing to alert at all would score perfectly on the one metric and be useless.

6.6 The obvious baseline, run against ourselves

Table 10 compares the cascade against the raw detector *at the same confidence floor*. That leaves open the objection any careful reader raises next: forget temporal validation — raise the floor until the false-alarm rate matches, and see what it costs. Most system papers leave this control to the reviewer. We ran it, and it partially deflates our own headline, so it is reported in full.

Sweeping the raw detector's floor over the same cached detections, a floor of 0.26 — selected *on these very negatives*, so an oracle baseline — reaches the cascade's false-alarm rate of 0.010 while keeping frame recall 0.900 against the cascade's 0.792, all 6/6 fires, and a median time-to-alert of 1.0 minute against the cascade's 3.0. A floor of 0.30 drives the false-alarm rate to zero on this set outright, at frame recall 0.842. Figure 9 shows the frontier; the cascade's operating point sits below it.

6.6.1 What the control does and does not establish

The control establishes that **per-frame false-alarm reduction on this negative set is not the cascade's contribution**: a tuned threshold does it more cheaply. Three things bound how far that conclusion travels, each a fact, none a defence:

- **The matched floor is tuned on the evaluation negatives.** The sweep selects $\theta = 0.26$ using the same 30 sequences it is scored on; the cascade runs the detector authors' published default (0.20) untouched. A deployed threshold must be chosen per camera network without access to its future negatives, and how well that transfers is exactly what a single negative set cannot measure.
- **No floor survives the hard negatives.** On the curated `pyro-sdis` negatives (section 5.1), raising the floor to 0.30 still leaves a false-alarm rate of 0.648 while recall is already falling — fog and dust are detected *with high confidence*. Whether persistence helps there is untested, because no temporal hard-negative set exists in any release we know of; building one is listed in section 12.2.

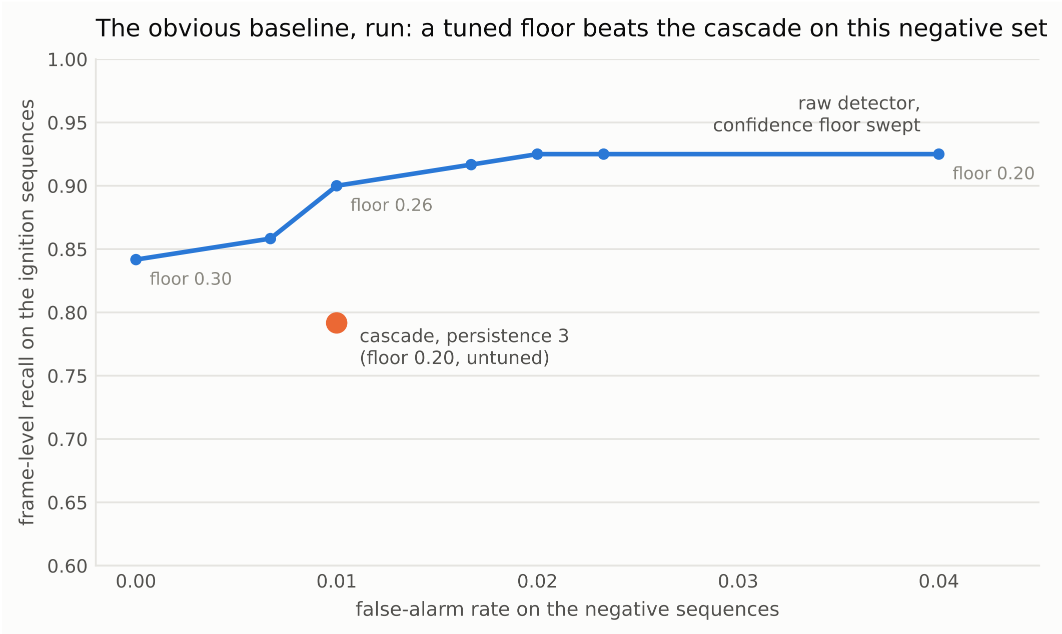

Figure 9. The control experiment (`eval/run_threshold_matched.py`). Each blue point is the raw detector at one confidence floor, on identical cached detections; the orange point is the cascade at persistence 3. On this negative set an oracle-tuned floor dominates the cascade — the finding this paper reports against itself, and the reason the claim in section 6.6.1 is scoped the way it is.

- **Deduplication is not achievable by any threshold.** Table 9 shows most of what the validator does on several hazards is withholding repeats of an event already reported — an operational function a per-frame threshold cannot express at all.

The honest statement of the validator's value is therefore narrower than an earlier draft of this work believed, and is the one this paper stands behind: *a single untuned mechanism that reaches an oracle-tuned threshold's false-alarm rate without access to the negative distribution, provides deduplication no threshold can, and transfers across hazards by configuration* — at a measured cost in recall and latency (section 6.7) that thresholding, where it suffices, does not pay. Where a deployment can tune a floor on representative negatives and does not need deduplication, thresholding alone is the better choice, and this paper says so.

6.7 The latency cost, in closed form

The latency is not an empirical surprise. Persistence requires N associated sightings before confirming, so for a camera sampling at rate f the additional delay relative to a single-frame detector is

$$\mathbb{E}[\Delta t] = \frac{N-1}{f}, \tag{16}$$

assuming the track is not missed. HPWREN's FIgLib cameras sample once per minute, so at $N = 3$ eq. (16) predicts exactly $(3-1)/1 = 2$ min of extra delay. The measured medians are 1.0 minutes raw and 3.0 validated, a difference of 2.0 minutes — the predicted value. The identity matters more than the agreement: it says the latency cost is a property of the *camera's sampling rate*, not of the method, so the same cascade on a camera sampling at 1 Hz costs two seconds. A deployment choosing N is choosing a point on eq. (16), and can be told what it will cost before any data is collected.

6.8 Hazard-agnosticism, tested and enforced

PyroNear's published training manifest shows that they already perform sequential confirmation for wildfire, with a grid search over a minimum frame count. Temporal confirmation for fire is therefore not novel, and we say so. The claim VIGÍA makes is narrower and still defensible: *one* validator, configured rather than specialised, working across hazards whose detectors were built by different people, on different data, for phenomena with different physics.

Table 11. The same validator on two hazards with different physics: a wildfire plume develops over tens of minutes and is nearly static between frames; a collision resolves in under a second with objects moving fast across the frame. The only difference between the two runs is a configuration object. MEASURED — `eval/run_multihazard.py`.

Hazard	Source	Frames	Raw det.	Confirmed	Suppression %
Wildfire smoke	FIgLib, 1 frame/min	240	109	15	86.2
Road incidents	Dashcam, 2 fps	660	291	33	88.7

ENFORCED, NOT MERELY ASSERTED

`tests/test_validator.py` contains `test_validator_contains_no_hazard_specific_code`, which parses the validator's source with Python's AST module and fails the build if any comparison against a specific hazard appears. A future contributor fixing a wildfire bug with a hazard-specific branch cannot silently invalidate the central claim of this document. The suite passes 9 of 9.

7 The anticipatory flood signal

Flood is the only hazard whose temporal logic does not rest solely on the validator. A water level has a trend, and the trend is the alert: knowing that a street is flooded is useful, but knowing that it is filling, and how fast, is what buys evacuation time.

Not a novel contribution, and read this before describing it as one. Choi et al. [5] target the same gap on the same infrastructure — anticipatory urban flood warning from existing CCTV, using Page–Hinkley change detection with short-term trend analysis producing an estimated time-to-threshold. That is the same capability, published independently. VIGÍA therefore implements *both* mechanisms and measures them against each other on identical inputs, rather than claiming either. What is ours is the multi-hazard pipeline they sit inside.

7.1 Two signals

Segmenting water yields a mask $W_t \subseteq \Omega$ over the frame domain Ω. Two scalars are derived, deliberately:

$$a_t = \frac{|W_t|}{|\Omega|} \qquad \text{area fraction — needs no calibration} \tag{17}$$

$$\ell_t = \frac{1}{K} \max\{\, k \;:\; \textstyle\sum_{j=k-4}^{k} \mathbb{1}[\, p_j \in W_t \,] > 0 \,\} \qquad \text{level along a reference line} \tag{18}$$

where $p_1, \dots, p_K$ are $K = 200$ points sampled from the bottom of a per-camera reference line upward. Equation (18) walks up from the bottom and stops at the first sustained dry stretch rather than taking the highest water pixel anywhere on the line; a reflection or a puddle further up would otherwise read as a far higher level than reality. The five-sample tolerance is 2.5 % of the line, which absorbs segmentation speckle without jumping gaps.

The area signal works on any camera the moment it is connected; the level signal needs a one-off human calibration but is physically meaningful and comparable between cameras. Choosing the reference line is deliberately not automated: a wrong automatic choice would silently corrupt every reading from that camera.

7.2 Windowed least squares

Rate of rise is fitted, not differenced. Frame-to-frame differencing of a noisy segmentation mask produces a rate estimate dominated by segmentation jitter; an ordinary least-squares slope over a window is stable enough to threshold on. Over the readings in a trailing window $\mathscr{W}$,

$$\hat{\beta} = \frac{\sum_{i\in\mathscr{W}}(t_i - \bar{t})(y_i - \bar{y})}{\sum_{i\in\mathscr{W}}(t_i - \bar{t})^2}, \qquad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{19}$$

A trend is declared only when $|\mathscr{W}| \geq 4$ and $R^2 \geq 0.5$: a poor fit usually means the segmentation is unstable rather than that the water is doing something complicated, so the system declines to call it. Where a calibrated level exists it is preferred over area for the trend decision, because area fraction moves with camera perspective and level does not. The operational output is the extrapolation

$$t_{\text{threshold}} = \frac{\ell^* - \ell_t}{\hat{\beta}}, \qquad \hat{\beta} > 0, \tag{20}$$

which is the number an operator actually wants: not “the street is flooding” but “this underpass is impassable in eleven minutes”.

A BUG ONLY REAL DATA COULD FIND

With a fixed 300 s window, eq. (19) never fired at all on a creek camera sampled every fifteen minutes — it reported UNKNOWN for seven hours while the water rose by 10 % of the frame — because the window could never contain two frames. Every synthetic test passed throughout. A time window is the wrong abstraction when the sampling rate is a property of the deployment, so the window is now adaptive: it admits at least four sampling intervals, measured from the observed median gap. Page–Hinkley, being sample-based, not time-based, was unaffected.

7.3 Page–Hinkley change detection

The Page–Hinkley test [13, 21] is a classical sequential change-point detector. For observations $x_1, x_2, \ldots$ with running mean $\bar{x}_i$, tolerated drift δ and threshold λ:

$$m_t = \sum_{i=1}^{t} \big(x_i - \bar{x}_i - \delta\big), \tag{21}$$

$$M_t = \min_{1\leq i\leq t} m_i, \tag{22}$$

$$\mathrm{PH}_t = m_t - M_t, \tag{23}$$

with an alarm at the first $t \geq t_{\min}$ for which $\mathrm{PH}_t > \lambda$. Its character differs from eq. (19): it responds quickly to the *onset* of a change, whereas the fit estimates the *rate* of an ongoing one more stably. They are complementary rather than competing, which is why both are available.

7.4 Calibration matters more than the algorithm

On a signal that is not changing, eq. (21) is a random walk with step standard deviation σ and drift $-\delta$ per sample, and eq. (23) is that walk's drawdown below its own running minimum. A threshold small relative to the noise will therefore be crossed eventually by noise alone. For Brownian motion with negative drift the all-time maximum drawdown is exponentially distributed, giving

$$\Pr[\text{false alarm}] \;\approx\; \exp\!\Big(-\frac{2\,\delta\,\lambda}{\sigma^2}\Big), \tag{24}$$

and, inverting, a design rule for a camera nobody has calibrated:

$$\lambda \gtrsim \frac{\sigma^2}{2\delta} \ln \frac{1}{\alpha} \tag{25}$$

for a target false-alarm probability α.

Our first defaults, $\delta = 0.002$ and $\lambda = 0.02$, looked reasonable and produced a 92 % false-alarm rate on *stable* water with realistic segmentation noise. Table 12 places the measured grid beside eq. (24).

Table 12. Page–Hinkley false-alarm rate on stable water, measured over 200 trials of 40 samples, against the drawdown model of eq. (24). All values are percentages. The model consistently *under*-predicts — by roughly a factor of two in the interesting region — because it assumes the infinite-time limit and treats the running-mean subtraction as exact. It nevertheless reproduces the ordering across every row and column and locates the usable/unusable boundary correctly, so eq. (25) is a sound floor for a new camera and not a substitute for measuring it. Generated by `docs/build_paper.py` from `eval/results/page_hinkley_calibration.json`.

δ	λ	$\sigma = 0.005$		$\sigma = 0.01$		$\sigma = 0.02$	
		obs.	model	obs.	model	obs.	model
0.002	0.02	15.0	4.1	92.0	44.9	100.0	81.9
0.002	0.05	0.0	0.0	23.0	13.5	90.5	60.7
0.005	0.05	0.0	0.0	2.0	0.7	74.5	28.7
0.005	0.10	0.0	0.0	0.0	0.0	16.0	8.2
0.010	0.10	0.0	0.0	0.0	0.0	2.0	0.7

The chosen default, $\delta = 0.005$ and $\lambda = 0.10$, trades a little latency for zero false alarms at typical segmentation noise: detection latency for a genuine rise is 11, 7 and 4 samples at 0.5 %, 1 % and 3 % per frame respectively. A camera with noisier segmentation needs a higher threshold, and eq. (25) says roughly how much higher — but it should be measured per camera, not assumed.

7.5 Compared on real cameras

Both mechanisms were run on identical inputs from LSU creek camera sequences in the V-FloodNet dataset [18]: real fixed cameras, real water, real segmentation noise. The two win on different sampling regimes — the windowed fit is faster on densely sampled cameras, Page–Hinkley on sparsely sampled ones, because it counts samples rather than seconds. Neither is claimed as ours, and the comparison is reported; no winner is declared.

7.6 Ground truth the other footage does not have

USGS cameras [27] sit on instrumented gauges, so every frame arrives with a measured water level beside it — which no other footage in this project provides. Over the 50 frames of a real flood event (Walnut Creek at Raleigh, 8 August 2024, gauge rising to 9.00 ft and receding to 4.25 ft), segmented water coverage correlates with measured gauge height at $r = +0.50$.

That is a real but moderate relationship, and it is reported because it *qualifies* the trend claim instead of supporting it unconditionally. Coverage tracks the water level between a flood day and a receded day, but within a day at near-constant level the coverage still varies by more than the level does. Restricting to midday frames does not improve it ($r = +0.46$ to $+0.50$ across daylight windows), so the residual is not simply low sun. Camera-derived coverage is evidence of a trend, not a substitute for a gauge. This is one event on one camera, and section 12.2 lists repeating it as outstanding work.

8 Scene routing on the upload path

Tier 0 decides which detectors run on a *registered* camera, from a declared context. Uploaded footage has no registration and no declaration, so something has to choose. This section describes that something, and is separated from section 3.1 deliberately: the two answer superficially similar questions under completely different evidence, and conflating them would undermine the argument of section 3.1.

8.1 Two approaches that did not work

Asking the detectors. The first version ran every detector over a few sampled frames and took whichever claimed the footage most strongly. It routed 2 of 4 test clips wrongly, and not for want of tuning: the scores are not comparable quantities. Each detector has its own confidence calibration and its own propensity to fire, so "flood scored 1.9 and building access scored 0.7" says almost nothing about which is right. On rubble the aerial flood model genuinely reports water — the known consequence of training on 51 flooded images (section 4.4).

Asking the user. The second version presented the detectors' reactions as ranked evidence and let the operator choose. Always correct, and one piece of homework for someone who came to watch a system work.

8.2 A classifier trained for this question

The third version is a MobileNetV3 classifier over 4 scene classes, trained on 1560 images and held out on 384. For a video it samples $K = 6$ frames and averages *probabilities*, not logits:

$$\bar{p} = \frac{1}{K}\sum_{k=1}^{K} \operatorname{softmax}\big(z(f_k)\big), \qquad \hat{c} = \arg\max_{c}\ \bar{p}_c. \tag{26}$$

Averaging after the softmax bounds each frame's influence at $1/K$. Averaging logits instead lets a single frame with an extreme activation — a close-up of water in an otherwise aerial rubble clip — dominate the whole decision, which is precisely the failure mode the detector-voting approach already exhibited.

Held-out accuracy is 1.000 (95 % interval [0.990, 1.000]), and per class 1.000 on 96 images each ([0.962, 1.000]). On the four demonstration clips it routes 4 of 4 correctly, at $\bar{p}_{\hat{c}}$ of 1.00, 0.98, 1.00 and 0.85.

8.3 How that figure should be read

An accuracy of 1.000 deserves suspicion before celebration, and three qualifications belong beside it.

- **The task is easy.** Landscape with a smoke plume, water seen across, open sea from above, and rubble from above are four visually distinct scene types. This is not fine-grained discrimination, and the number should not be read as though it were.
- **The model has memorised its training set**, reaching a training loss of 10^{-4}. Held-out accuracy is what matters, and the splits are source-aware — fire trains on `pyro-sdis` stills and tests on HPWREN sequences from other cameras; flood trains on FloodNet UAV imagery plus ATLANTIS ground photography and tests on USGS and LSU cameras it has never seen — but building access is split by capture index within one archive, which is the weakest of the four splits and should be read as such.
- **Road incidents are absent.** Only four usable training images exist for that class, so the router was never taught it and never offers it. Excluding a class the evidence cannot support is preferable to a fifth output trained on four pictures.

The split that was wrong, and how it showed

The first trained router scored 1.000 on three classes and 0.60 on flood. The cause was a viewpoint mismatch of exactly the kind section 3.2 exists to prevent: the flood class trained on FloodNet, which is entirely aerial, and was tested against ground-level cameras. Water seen from above and water seen across are different pictures. Adding ATLANTIS ground-level photography to that one class moved it to 1.000. The number without this story looks like luck.

8.4 This does not overturn Tier 0

The router is confined to the upload path. A registered camera's hazards are still *declared*, never inferred, so the argument of section 3.1 — that a misrouting classifier on the critical path fails silently — is untouched. The router operates where nothing has been declared and where a human is present to disagree: its answer is applied automatically, stated in the interface, and every alternative remains one click away. That is a different risk profile from a live camera watching unattended, and the architecture keeps the two apart.

9 Evaluation methodology

9.1 Commitments

- **Cross-dataset evaluation.** Domain shift is the most likely cause of real-world failure; a random split of the training set flatters a model and predicts nothing about deployment.
- **Report the gap.** In-domain and out-of-domain figures appear side by side.
- **Event-level evaluation on sequences**, not frame-level on stills. The validator only demonstrates value over time.
- F_2 **rather than** F_1 **as the headline**, for the reason given in eq. (6).
- **Never quote a figure we did not measure.** Published third-party numbers appear only in clearly attributed rows.
- **Every proportion carries its interval** (section 4.5).

A TRAP WE WALKED INTO AND DOCUMENTED

Our first false-alarm measurement on ignition sequences gave 0.8 %, against 71 % on curated hard negatives with the same model and threshold. The difference is negative-set difficulty: frames shortly before a fire are usually clear skies, whereas curated negatives are cloud, fog and dust chosen to confuse. Had we published the 0.8 % figure as evidence the validator works, any reviewer inspecting the negative set would have dismantled the claim. Every false-alarm figure in this document names the negative set it came from.

9.2 Reproducibility

Detection is run once and cached to JSONL; validator configurations are then replayed against the cache. Every configuration therefore sees byte-identical detector output, so any difference in outcome is attributable to the validator and not to run-to-run variation. All results files are committed under `eval/results/`, and every figure in this document is generated from them (appendix A).

A third guard, `tools/freeze_results.py`, checks each published figure against the results file it cites and fails the build if the two disagree.

THE FREEZE GUARD CATCHING ITS AUTHOR

The aerial flood model was first registered with its figures marked `measured`, the same label the held-out evaluations carry, with no source file named. The guard rejected it. The entry now reads `validation_measured` with an explicit note that FloodNet's withheld masks make an independent split impossible — which is the distinction section 5.2 now draws in the text as well.

9.3 Execution backends

Two backends exist and are not interchangeable. REFERENCE runs the as-published dynamic-shape graph on the CPU provider and is deterministic; every figure in this document comes from it. FAST runs a static-shape variant on Apple's CoreML provider and is 1.7–1.9× faster (34 ms against 60 ms at 1 024 px).

FAST is not bit-identical. Across 100 validation frames the detection count matched on 100 of 100 and boxes agreed to within 0.41 px, but confidences drift by up to 0.012 because the Neural Engine computes at lower precision. At the 0.20 operating point that drift changed no metric; at 0.10 and 0.30 it moved box precision and recall by about one percentage point by flipping a single detection across the threshold. FAST is therefore used for anything a person watches, and REFERENCE for anything reported.

A PARITY CHECK THAT CHECKED NOTHING

The ONNX export parity check originally traced the model on random noise. A trained detector finds nothing in random noise, so the check compared zero detections against zero detections and passed unconditionally. It now traces a real frame and reports UNVERIFIED if that frame yields no detections, instead of passing quietly.

9.4 Demonstration material

Demonstration clips are treated as evidence, not decoration, and are subject to the same gating as any published figure. One clip per hazard is assembled from local data in sorted order at a fixed frame rate, so a clip built twice is identical and a run over it reproduces exactly. The recording is rendered directly from the pipeline instead of screen-captured, which removes any dependence on a browser, a display or a network at presentation time.

Two independent conditions decide whether a clip may appear in a recorded demonstration, a figure or any public material. Both are recorded per clip in a manifest, and the renderer refuses a clip that fails either — refusing is the safe default, because a clip that should not be shown plays exactly like one that may.

1. **The licence permits publication.** The source datasets carry four different licences and one forbids redistribution outright. Where a licence permits publication subject to attribution, the credit is burned onto every frame not left to a caption that may be cropped away.
2. **The frames are a genuine time sequence.** The validator is temporal, so demonstrating it on footage without temporal continuity misrepresents it in both directions: persistence rejects almost everything because nothing genuinely persists, while a few detections confirm anyway because unrelated objects happen to overlap between consecutive frames and the tracker associates them.

A CLIP THAT PLAYED PERFECTLY AND MEANT NOTHING

The first people-in-water clip was 60 unrelated validation stills stitched into a video. It played correctly and produced confident figures — 156 detections proposed, 16 confirmed, 121 rejected by persistence — all of them artefacts. Continuity is now verified, not assumed: adjacent-frame histogram correlation is about 0.997 on genuine footage against about 0.015 for unrelated frames from the same split, and the fetch script refuses to report success unless that separation holds. On real sequential footage the same detector and validator give 451 proposed and 59 confirmed.

A SOUND METHOD ON AN UNREPRESENTATIVE SAMPLE

DRespNeT is published as shuffled, renamed stills whose filenames retain the original capture index, and ordering by that index does recover the authors' order. A twelve-frame window looked continuous — adjacent correlation 0.695 against 0.235 for unrelated pairs — and a 66-frame clip was built on that basis and described as a genuine sequence. That was wrong. The authors also publish their un-augmented raw frames, and across all 615 of them the median adjacent correlation is 0.388 with 362 scene cuts; the longest continuous run in the entire dataset is *seven* frames, and within the 66-frame range actually used, 40 of 65 transitions are scene cuts. The check was the right kind of check — asking whether detections associate across frames rather than whether frames look alike — which is what made the conclusion feel well-founded. Tracks did reach length seven and 124 associations survived to persistence three, but on footage that is 62 % scene cuts those associations are coincidental spatial overlap between unrelated scenes. The claim was retracted; the clip is retained for exercising the detector only. Sample size is the part that has to be checked first.

Table 13. Demonstration footage and its licence. Every clip in the demonstration path is publicly licensed; no restricted footage remains.

Hazard	Source	Licence
Fire	HPWREN FIgLib ignition sequence [15]	attribution required
Flood	USGS gauge camera, real flood event [27]	public domain
People in water	SeaDronesSee flight sequence [28]	CC0 1.0
Building access	DRespNeT stills — *no sequence exists* [26]	CC BY 4.0

Flood was solved by changing source, not by relaxing the rule. The USGS operates over 1,300 streamgage cameras whose imagery is unrestricted, and they are the right kind of footage as well as the right licence: fixed mount, pointed at water, sampled roughly hourly — the same sparse regime the trend detection was calibrated for, and the regime in which a fixed 300 s window originally failed outright (section 7). Only daylight frames are used: these cameras switch to infrared at night, and on night frames the segmenter tints sky and vegetation as water. That is the detector's declared envelope, and the fetch script enforces it instead of restating it.

10 Consolidated results

Every figure VIGÍA has measured to date, in one place. Figures published by third parties are listed separately in table 16 and are never combined with ours.

11 Attribution and credits

VIGÍA is built on work done by other people. This section names them. Model weights are not redistributed in our repository; they are downloaded from their original distributors at setup time, and provenance for each is recorded in `models/REGISTRY.yaml` alongside its licence.

11.1 Prior art

VIGÍA is not the first camera-based hazard detection system and does not claim to be. The following work precedes it and in several respects exceeds it.

11.2 Runtime dependencies

The runtime is four packages: `onnxruntime` (MIT), `numpy` (BSD-3-Clause), `opencv-python-headless` (Apache-2.0) and `PyYAML` (MIT). The `ultralytics` package (AGPL-3.0) is deliberately not a runtime

Table 14. Measured by us. Every value is generated from the results file named in the last column; none is typed into this document.

Hazard	Metric	Value	Harness
Fire	Box precision	0.447	`run_image_eval`
Fire	Box recall	0.902	`run_image_eval`
Fire	Image-level FAR, 585 negatives	0.706	`run_image_eval`
Fire	FAR, no validator	0.040	`run_ablation`
Fire	FAR, validator at $N = 3$	0.010	`run_ablation`
Flood	Water IoU, ATLANTIS test	0.7633	`run_flood_eval`
Flood	F_1, ATLANTIS test	0.8657	`run_flood_eval`
Flood	Water IoU, aerial, flooded subset	0.4734	`train_flood_aerial`
People in water	Box F_1	0.9172	`run_person_in_water_eval`
People in water	Image-level FAR	0.0222	`run_person_in_water_eval`
Building access	Civilian box recall	0.942	`run_building_access_eval`
Building access	Civilian box F_1	0.856	`run_building_access_eval`
Scene routing	Held-out accuracy	1.000	`train_scene_router`

Table 15. Quantities that are not detector accuracy: what the validator does, what the gate saves, and what the system costs to run. MEASURED on the reference machine (MacBook Pro M4 Pro, 48 GB), REFERENCE backend.

Quantity	Value	Source
False-alarm reduction, validator at $N = 3$	75 %	table 10
Ignition sequences still detected	6/6	table 10
Median time to first alert, raw	1.0 min	table 10
Median time to first alert, validated	3.0 min	table 10
Added latency, predicted and observed	2.0 min	eq. (16)
Frame-level recall, raw then validated	0.925 / 0.792	table 10
Threshold-matched control: oracle floor	0.26 reaches FAR 0.010	section 6.6
its recall / latency vs the cascade's	0.900 / 1.0 min vs 0.792 / 3.0 min	fig. 9
Colour prior: measured and removed	recall 0.978 → 0.736 for FAR 0.706 → 0.456	section 6.3
Suppression, total across four clips	91 %	eq. (15)
of which filtering	74.4 %	table 9
of which deduplication	16.3 %	table 9
Gate reduction, demonstration registry	80 %	eq. (1)
detector invocations per frame	8 of 40	eq. (1)
Inference latency, REFERENCE	~60 ms	section 9.3
Inference latency, FAST	~34 ms	section 9.3
Validator behavioural tests	9/9 passing	`tests/test_validator.py`
Registry tests	22/22 passing	`tests/test_registry.py`
Flood level tests	13/13 passing	`tests/test_flood_level.py`

Table 16. REPORTED BY AUTHORS. Published by the people who built each component, listed here for completeness and never combined with our figures. Where a model card publishes nothing, that is stated, not filled in.

Component	Metric	Value
PyroNear fire detector	precision / recall / mAP	not published
PyroNear2025 benchmark	cross-dataset F_1	~70 %
RF-DETR SeaDronesSee	mAP@50, five classes	0.7931
RF-DETR SeaDronesSee	per-class AP, swimmer	0.282
YOLOv8-DRN (DRespNeT)	mAP@50, 28 classes, 27 FPS	92.7 %
Road-incident YOLO11x	mAP@50	0.826
Road-incident YOLO11x	accident-class recall	0.855
Reference multi-task system	fire FAR 52 % → 4 % at 96 % sensitivity	[1]

Table 17. Models and datasets, with the licence each is used under.

Component	Licence	Used for
`yolo11s_sensitive-detector` — PyroNear association with the French Fire and Rescue Services (SDIS)	Apache-2.0	fire detection
`pyro-sdis` dataset — PyroNear volunteers with SDIS partners	Apache-2.0	fire evaluation
`pyro-engine` — PyroNear association	Apache-2.0	architectural reference only; no code copied
FIgLib and camera archive — HPWREN, UC San Diego	attribution required	ignition sequences, negatives, live cameras
ATLANTIS — Erfani et al., iWERS lab, University of South Carolina	CC / public-domain source images; annotation status unstated	flood training and test
FloodNet — Rahnemoonfar et al.	CDLA-Permissive-1.0	aerial flood training
SeaDronesSee — Varga, Kiefer et al., University of Tübingen	CC0 1.0 (data), MIT (code)	people-in-water detection
`seadronessee-rfdetr-small` — dronefreak	Apache-2.0	people-in-water detection
DRespNeT / YOLOv8-DRN — Cranfield University	CC BY 4.0	building-access detection
`traffic-accident-detection-yolo11x` — Enos-123	MIT	road-incident detection
V-FloodNet LSU sequences — Liang et al.	evaluation use only	flood trend comparison
USGS HIVIS streamgage cameras	public domain	flood footage and live cameras
torchvision (DeepLabV3, MobileNetV3) — Meta AI	BSD-3-Clause	flood architecture

Table 18. Prior and concurrent work.

Organisation	Domain	Note
Pano AI	wildfire	Commercial; ~USD 50,000 per camera per year including a 24/7 intelligence centre
ALERTCalifornia / DigitalPath	wildfire	UC San Diego and CAL FIRE; 1,060+ cameras with human vetting
PyroNear	wildfire	Open-source, deployed across France, Spain and Chile — and the direct source of our fire detector
Lynxight, AngelEye, Coral	drowning	Commercially mature; ASTM F3698-24 and ISO 20380:2017
Rekor, Iteris, Miovision, NoTraffic	road	Deployed with US state transport departments
xView2 / DIUx [11]	building damage	Satellite damage assessment; needs bi-temporal pairs including a pre-disaster image, so not a camera model
RipVIS [25]	rip currents	CVPR 2025 benchmark; source of the F_2 convention used here
Choi et al. [5]	urban flood	Page–Hinkley for anticipatory CCTV flood warning — the same capability as section 7.3, published independently

dependency and is confined to an isolated build environment (section 3.4). The web interface is an optional extra: a deployed edge node runs headless and never needs it.

12 Conclusion

VIGÍA set out to make published hazard detectors trustworthy enough for the cameras cities already own. What survives its own evaluation is narrower and better-founded than what an earlier draft believed, and stating the final form plainly is the point of this section.

The system stands: five detectors from four independent sources run behind one declarative gate with enforced viewpoint envelopes, one temporal validator confirmed to contain no hazard-specific branch, a privacy boundary with no biometric code path, and an AGPL-free runtime of four packages — each property held by a build-failing guard, not a promise. The validator's value, after the controls of section 6.6, is scoped to what the measurements support: it reaches an oracle-tuned threshold's false-alarm rate *without* access to the negative distribution, it deduplicates alerts as no per-frame threshold can, and it transfers across hazards by configuration — at a closed-form latency cost (eq. (16)) and a measured recall cost that thresholding, where it suffices, does not pay. Two cascade levels that failed their measurements were removed, not kept (section 6.3).

12.1 Findings that travel

Distilled from the measurements, phrased for reuse beyond this system:

1. **Report suppression as two numbers or not at all.** Filtering and deduplication differ by a factor of eight between our hazards (table 9); one figure conflates a false-alarm claim with alert housekeeping.
2. **Run the threshold-matched control before claiming alarm reduction.** On easy negatives it beat our cascade; on hard negatives nothing beats anything yet. Either result changes what may be claimed, and it costs one afternoon on cached detections.
3. **Persistence has a price list:** $\mathbb{E}[\Delta t] = (N-1)/f$. The latency cost of temporal confirmation is a property of the camera's sampling rate, computable before any data is collected.
4. **Pooled metrics flatter minority-critical classes.** Selecting the aerial flood model on pooled IoU would have chosen a model better at recognising dry ground (section 4.4); stratify on the case the system exists for.

5. **Appearance priors fail where the confuser shares the property** — fog is as grey as smoke — and geometric priors can invert safety on the severe tail (the worst floods are the largest). Measure each level of a life-safety cascade on the negatives it will actually face, and be willing to delete.
6. **Image-level metrics can hide a dead detector.** A label-shift bug survived image-level evaluation at precision 0.83 and was exposed only at box level, at recall 0.002 (section 4.2). Evaluate at both levels, always.
7. **Interval or it didn't happen.** At $n = 12$ images, a recall of 1.000 means $[0.757, 1.000]$ (table 3); reporting the interval is the difference between a measurement and an anecdote.

We would argue the transferable contribution is the practice: every proportion carrying the interval its sample supports, stratified metrics where pooling flatters, suppression decomposed into the claim it supports and the housekeeping it is not, controls run against one's own headline, and twelve retracted claims documented in place. None of this required novel architecture. All of it changed what the paper was allowed to say.

12.2 Outstanding work

Listed in the order it would be tackled. The first three are shortages of available material rather than defects in the system.

1. **No temporal hard-negative set exists.** Tuned thresholding suffices on easy negatives (section 6.6); no threshold survives the curated hard stills; whether persistence survives them is untestable without hard negatives that are also sequences — fog rolling over a ridge, filmed. Assembling one from HPWREN's archive is the single most valuable experiment this project has not run.
2. **Building access rests on 24 held-out images.** Strong on that split (civilian box recall 0.942, interval [0.908, 0.963] over instances), but twelve positive images cannot make the image-level figures precise (table 3), and the released data cannot be widened. It needs a different post-disaster aerial dataset with people annotated.
3. **No urban flood footage.** The flood demonstration uses a public-domain gauge camera on a creek. The 2024 Valencia DANA is the event this system is motivated by, and no footage of a flooded street is yet available to it under a licence permitting publication.
4. **The road-incident detector has no standalone measurement** against held-out labels; it remains the validator's generality experiment only.
5. **The gauge correlation is one event on one camera** ($r = +0.50$, section 7.6); it should be repeated across many USGS cameras and events before the trend claim is leaned on.
6. **The aerial flood segmenter has no held-out split** (table 6); FloodNet's withheld masks make this unfixable from the public release.
7. **Night operation.** Fire and flood are evaluated in daylight only; the flood segmenter mislabels infrared frames and the fetch pipeline excludes them by envelope. Disasters do not keep daylight hours — the Valencia flooding peaked in the evening — so an infrared-domain evaluation, likely fine-tuned on the USGS cameras' own night frames, is required before any claim extends past dusk.

Data and code availability. Every figure in this paper is generated from committed measurement files by the build described in appendix A; the repository, including the training and evaluation scripts that reproduce each number, accompanies this paper.

Acknowledgments. This work runs on other people's science, named throughout and in section 11: the PyroNear association and the French SDIS; HPWREN at UC San Diego, whose imagery appears here under its attribution requirement; the U.S. Geological Survey's streamgage camera network; the authors of ATLANTIS, FloodNet, SeaDronesSee, DRespNeT and V-FloodNet; and the maintainers of ONNX Runtime, NumPy, OpenCV and torchvision.

A Reproducibility

A.1 How this document is built

No figure in this document is typed. `docs/build_paper.py` reads `eval/results/*.json`, regenerates every chart from the same files, writes `generated/numbers.tex` as a set of macros, and only then runs the typesetter. A measurement therefore changes this document by rebuilding it. Two derived quantities — the Wilson intervals of eq. (9) and the Page–Hinkley model of eq. (24) — are computed at build time rather than stored, because they are properties of the measurements rather than new measurements.

```
make paper        # figures -> numbers -> LaTeX -> vigia.pdf
make check        # licence guard, privacy guard, results freeze, tests
make dev          # the whole demonstration, one command
```

A.2 The three guards

Table 19. Build-failing guards. Each turns a commitment that would otherwise live in prose into one that cannot be broken silently.

Guard	What it prevents
`tools/check_licence.py`	Any import of an AGPL-licensed package anywhere under `vigia/`. Parses the AST instead of grepping, so a comment mentioning the package does not trip it (section 3.4).
`tools/check_privacy.py`	Any biometric identification code path across `vigia/`, `tools/`, `scripts/` and `eval/` (section 3.5).
`tools/freeze_results.py`	Any published figure that disagrees with the results file it cites (section 9).

A.3 Reference hardware

All figures in this document were produced on a MacBook Pro (M4 Pro, 48 GB) using the REFERENCE backend: the as-published dynamic-shape ONNX graph on the CPU execution provider, which is deterministic. Section 9.3 describes the faster non-deterministic alternative and why it is never used for a reported number.

A.4 Data acquisition

None of the datasets used here are redistributed. Each is fetched from its original distributor by a script under `eval/`, which records the licence and refuses to report success if the fetched material fails the continuity or licence conditions of section 9.4. Acquisition routes for material not yet obtained are recorded in `docs/DATA_ACQUISITION.md`.

B Revision history

Every entry that adds or revises a measured figure is recorded here, so the provenance of each number in this document can be traced.

Table 20. Revision history. Dates are 2026.

Ver.	Date	Change
0.1	1 Sep	Project foundation: ONNX-only runtime, licence guard, model registry, fetcher. Fire detector integrated and first baseline measured.
0.2	1 Sep	Temporal validator implemented with five levels and an IoU tracker. Ablation measured: 75 % false-alarm reduction at persistence 3 with no loss of fires detected. Per-level ablation showed size plausibility contributes nothing.
0.3	1 Sep	Road-incident detector integrated via isolated ONNX export. Hazard-agnostic transfer measured across two hazards and enforced by an AST-based test. Scope revised: building access and people-in-water added, pool drowning removed.
0.4	1 Sep	ATLANTIS acquired; class mapping derived empirically from the masks. Water level and rate-of-rise implemented and tested. Size-plausibility level removed after measurement across three hazards; all prior results unchanged, confirming it contributed nothing.
0.5	1 Sep	All four hazards measured. Flood segmenter trained on ATLANTIS. People-in-water integrated, where a label-shift bug reported boats as swimmers while discarding real swimmers — caught only by box-level evaluation. Building access trained on DRespNeT.
0.6	2 Sep	The system became a system: context gate, runtime pipeline, egress boundary and operator view, with privacy enforced by a build-failing guard. Demonstration recorded deterministically for all four hazards, every clip publicly licensed. Results freeze added. Fire baseline re-measured on 585 negatives instead of 23: the false-alarm rate fell from 0.870 to 0.706 and every quotation was corrected. Flood footage moved to public-domain USGS gauge cameras.
0.7	2 Sep	Building access rebuilt on the DRespNeT authors' own published weights. Civilian box recall moved from 0.141 to 0.942, turning the weakest component in the system into one of the strongest. The claim that building access had a usable temporal sequence was retracted.
0.8	3 Sep	Aerial flood segmenter trained on FloodNet and routed by viewpoint, with the stratified/pooled distinction of section 4.4 made explicit. Scene router trained for the upload path (section 8).
0.9	3 Sep	Live public cameras integrated through the same pipeline (section 3.8). A blocking-endpoint defect was found and fixed: analysing a clip had stalled every live stream in the process for the duration.
1.0	3 Sep	Documentation rebuilt in LaTeX with the mathematics stated explicitly (section 4), interval estimation added to every proportion, the persistence-latency identity derived (eq. (16)), and the Page–Hinkley calibration compared against a closed-form model (table 12). Every figure is now generated from the results files rather than typed.
1.1	4 Sep	The two remaining evidence gaps in the cascade were closed by measurement, and both cut against the earlier draft. The threshold-matched control (section 6.6) showed an oracle-tuned confidence floor dominating the cascade on the FIgLib negative set; the headline claim was re-scoped accordingly, in the abstract and everywhere else. The colour prior — the one level never active in any reported experiment — was measured across five cutoffs on stills and sequences, found to trade recall for false-alarm rate at every point because fog and smoke are both grey, and removed like the size level before it (section 6.3), with a regression test barring its return.

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